

SoftwareX

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research --Manuscript Draft--

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Abstract:	Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant's personal history, current context, and the type of energy signal being delivered. This study presents BuildOcc, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset covering 16,684 respondents. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface - Python library, REST API, and Model Context Protocol server - so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers show that ATUS-grounded sampling reproduces empirically calibrated activity distributions and that demographic priors propagate into persona-consistent agent reasoning across timesteps, establishing internal consistency across strata. BuildOcc provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer. BuildOcc is openly released at https://doi.org/10.5281/zenodo.21192895 under the Apache License 2.0 and installable via <code>pip install buildocc</code>.

Response to Reviewers

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research

Manuscript SOFTX-D-26-00798 — SoftwareX

Wooyoung Jung

August 12, 2026

Dear Editor and Reviewers,

The author is grateful to the Editor and the two reviewers for their careful and constructive comments, which have improved both the accuracy of the manuscript’s claims and its usefulness to prospective adopters of the software. The manuscript has been restructured to the SoftwareX template, its quantitative claims have been qualified where the evidence does not support their earlier strength, and new material has been added on computational cost and on transferability beyond the United States. In what follows, each comment is shown in *italics*, the response as bullet points, and the corresponding revised manuscript text in *blue*. Cross-references are to sections and tables of the revised manuscript; exact line numbers will be supplied against the line-numbered proof.

Editor

Comment 1. *We ask that you follow our template for the heading sections (you have an extra section 4). You are free to add subsections as needed.*

Response.

- The manuscript now follows the five section headings prescribed by the SoftwareX template: Motivation and Significance, Software Description, Illustrative Examples, Impact, and Conclusions.
- The former standalone “Validation” section — the extra Section 4 — has been folded into Illustrative Examples, where the two validation tiers now appear as Sections 3.2 and 3.3 and the original walkthrough as Section 3.1. No content was removed in this restructuring.
- The activity-distribution figure, which the submitted version introduced in one section and discussed in the next, has been moved to sit with the Tier 1 discussion it supports. The two-tier framing paragraph that opened the former Validation section now introduces Section 3 as a whole.

Revised text (new structure of Section 3):

3. Illustrative Examples — 3.1 Demand response walkthrough; 3.2 Tier 1: Activity Schedule Fidelity; 3.3 Tier 2: Persona-Driven Behavioral Differentiation. Sections 4 (Impact) and 5 (Conclusions) follow.

Comment 2. *Also check that the word count is close (or less than) 4000 words.*

Response.

- The revision is 4,227 words in the main body (Sections 1–5), against 3,973 in the submitted version.
- The increase is concentrated in Section 2, which grew by 324 words to accommodate material the reviewers requested: the computational cost, runtime and multi-occupant scalability discussion (new Section 2.7; Reviewer 1 Comment 7 and Reviewer 2 Comment 4), the transferability discussion (Reviewer 2 Comment 5), the clarification of what temperature-zero reproducibility does and does not guarantee (Reviewer 1 Comment 3), and the primary-activity limitation (Reviewer 1 Comment 6).
- The sections Reviewer 2 asked to shorten were shortened. Illustrative Examples, which now also contains the former Validation section, is 117 words shorter than those two sections were at submission, despite absorbing the statistical and realism qualifications requested in Reviewer 1 Comments 1, 2 and 5 and Reviewer 2 Comment 2.
- Impact is essentially unchanged in length (367 to 380 words). The template identifies it as the main section, so it was reworded rather than cut. The Conclusions grew by 39 words, which is the limitation statement requested in Reviewer 1 Comments 4 and 6.
- The full accounting is therefore $3,973 + 324 - 117 + 13 + 39 - 5 = 4,227$: the remaining overage is the net cost of the requested additions. A note explaining this has been provided to the publication office with the resubmission.

Reviewer #1

This is an important development with strong background. However, there are few technical concerns need to be addressed.

Comment 1. *The second validation test only uses 3 to 5 runs per group. Saying “40% accepted” when that means 2 out of 5 people sounds a lot more solid than it is. Please clarify this.*

Response.

- This concern is well founded, and the presentation has been revised accordingly: percentages have been removed from the signal-compliance results. Table 8 now reports the number of accepting agents out of five seeds — for example 2/5 rather than 40% — and the caption states why percentages are not given.
- The surrounding text in Section 3.3 and the two figures quoted in Section 4 have been converted to the same counts, so no percentage derived from five runs remains anywhere in the manuscript.
- A closing sentence makes the epistemic status explicit.

Revised text (Table 8 caption):

Signal compliance per stratum and signal type, reported as the number of accepting agents out of five random seeds. Percentages are not given, since five runs per cell cannot support them.

Revised text (Section 3.3):

Table 8 reports the number of accepting agents per stratum and signal type. Direct commands (Type A) drew the most acceptance for O2 (retired couple, 2 of 5 seeds), with O1 (employed single) and O4 (unemployed adult) accepting in 1 of 5; educational price signals (Type B)

were accepted by O2 and O4 in 2 of 5 seeds each, a pattern consistent with cost-aware strata responding to both authority and economic framing. O3 (employed parent) rejected all signal types across all five seeds: signal delivery at 18:30 coincides with child-care activity codes in the ATUS schedule, making that outcome an expected consequence of schedule grounding rather than a general insensitivity to demand response. Social norm signals (Type C) drew no acceptance in any stratum, consistent with field evidence on the limited efficacy of peer-comparison messaging (Gyamfi and Krumdieck 2011; Albadi and El-Saadany 2008). With five runs per cell these counts indicate direction only; no significance testing is performed on them.

Comment 2. *There’s no real statistics run on these numbers anywhere. The author admits this, but it is mentioned as proven facts. Suggest mentioning that.*

Response.

- The claim that persona variation is “the primary driver” of the observed differences has been removed. The text now states only what the design supports: because the environment is held fixed, differences are attributable to persona rather than environment variation.
- An explicit statement that no significance testing is performed has been added where the counts are reported, and the verb “confirm” has been replaced by “show” wherever Tier 2 results are summarized. It is retained for Tier 1, whose divergence comparison over 180 simulated days per stratum does support it.
- The bound already noted in the submitted version — that the design is sufficient for directional patterns but not for statistical significance — is retained and now sits within a three-part statement of what limits the interpretation.

Revised text (Section 3.3):

Tier 2 tests whether the demographic priors encoded in each stratum’s persona produce distinguishable behavioral patterns under identical environment conditions, so that any observed differences are attributable to persona variation rather than environment variation.

Comment 3. *The paper says results are “reproducible” because temperature is set to 0. But different AI providers don’t all give the same steady answer at temperature 0. Please clarify this and add further explanation.*

Response.

- This is an accurate observation; the submitted manuscript left the scope of the reproducibility claim implicit. A paragraph has been added to Section 2.4 stating what $T=0$ does and does not guarantee.
- Determinism holds within a fixed provider, model version, and prompt. It does not extend across providers, or across successive versions of a hosted model, because it is relative to one specific set of model weights; provider-side updates can therefore alter behavior over time.
- Two practical consequences are stated: runs intended to be reproducible should record the provider and model version alongside the seed, and a locally hosted open-weight model can be pinned where long-term reproducibility matters.

Revised text (Section 2.4):

Setting the sampling temperature to $T=0$ makes the model return its single most probable output rather than sampling among alternatives, so a fixed provider, model version, and prompt reproduce the same action on every run. This determinism does not extend across providers, or across successive versions of a hosted model, because it is relative to one specific set of model weights. Reproducible studies therefore record the provider and model version alongside the seed, and a locally hosted open-weight model can be pinned where results must remain reproducible over time.

Comment 4. *The AI decides how important its own memories are, then uses that same rating to decide what to remember later. However, there is no feedback loop to ensure that the results become unstable.*

Response.

- The circularity identified here is acknowledged, and is now named explicitly as a limitation rather than left to be inferred from the appendix. The importance score is assigned by the agent in the same response that produces the action, and the same score then governs retrieval.
- The Conclusions state that this score is not externally calibrated and that no feedback path exists that would correct a mis-scored entry.
- The mechanism itself is unchanged in this revision, since altering the retrieval rule would change every reported result; it is instead documented as a known property of the architecture, which follows Park et al. in this respect.

Revised text (Section 5):

Current limitations include a single-action constraint per timestep; ATUS-grounded strata covering US demographic profiles only; activity grounding restricted to primary activities, since ATUS records only one activity at a time; and a memory-importance score that the agent assigns to itself and then reuses for retrieval, without external calibration or a feedback path that would correct a mis-scored entry. Subsequent phases extend the platform with multi-agent household simulation, non-US occupancy data, and external behavioral benchmarks including ecobee Smart Thermostat field data (U.S. Department of Energy 2021) matched to the four ATUS strata.

Comment 5. *One of the four groups, the unemployed adults, is based on only 107 people. The author admits this data is shaky, but still makes specific claims about how this group behaves. This should be accepted with a bit of caution.*

Response.

- The caution is now attached to the claim itself rather than only to the appendix that reports the sample size. The O4 interpretation is marked provisional at the point where it is made.
- The text states that the pattern may reflect sampling noise in the grounding data as much as a demographic tendency, and repeats the sample size inline so the reader does not have to consult Appendix C to weigh the claim.

Revised text (Section 3.3):

Because O4’s probability tables derive from only 107 ATUS respondents—the smallest of the four strata and the noisiest at 15-minute resolution—this pattern should be read as provisional; it may reflect sampling noise in the grounding data as much as a demographic tendency.

Comment 6. *As ATUS data only consider primary activities, it should be mentioned the secondary activities are a limitation of this model.*

Response.

- This limitation is acknowledged and is now stated in two places: where the grounding method is described, and in the Conclusions.
- Section 2.2 notes that ATUS records a single primary activity per episode, so simultaneous secondary activities are not represented in the scheduler’s output.

Revised text (Section 2.2):

ATUS records a single primary activity per episode, so simultaneous secondary activities—watching television while preparing food, for example—are not represented in the scheduler’s output.

Comment 7. *Calling the AI at every single timestep costs money. The author says cost is exactly why they couldn’t run more days or more test runs. That’s a real problem if someone wants to use this tool for a bigger study.*

Response.

- This is a fair concern for anyone considering adopting the software, and the submitted version mentioned cost only in passing. A new subsection, Section 2.7, now reports the cost structure directly: 96 model calls per simulated agent-day at the 15-minute timestep, with responses capped at 512 tokens, so monetary cost follows from the provider’s per-token price at the time of use; wall-clock time is dominated by provider latency rather than local computation.
- Two mitigations already available in the released version are described: any local open-weight model served through Ollama removes per-call cost entirely, and scheduler-only studies, including the Tier 1 validation reported here, require no model calls at all.
- The subsection states plainly that cost, rather than platform capability, is what bounded the simulation lengths reported in Section 3.

Revised text (Section 2.7):

Because the agent queries a language model at nearly every timestep, cost and runtime scale linearly with simulated agent-days. A 15-minute timestep gives up to 96 calls per simulated day per agent, plus occasional reflection calls, with responses capped at 512 tokens; the monetary cost therefore follows directly from the provider’s per-token price at the time of use, and wall-clock time is dominated by provider latency rather than local computation. Two mechanisms reduce this. Any local open-weight model served through Ollama removes per-call cost entirely, at the price of reduced reasoning quality, and scheduler-only studies—including the Tier 1 validation reported in Section 3.2—require no model calls at all. Cost, rather than platform capability, is what bounded the simulation lengths reported in Section 3.

Comment 8. *The paper shows a huge improvement, up to 1200 times better, over a basic fixed-*

schedule model. I agree with the improvement, however the 1200 times better seems to be highly overestimated. Suggest rephrasing this claim.

Response.

- This objection is well founded, and the claim has accordingly been withdrawn. The fold-reduction sentence has been removed from Section 3.2, and the corresponding ratio annotations have been removed from Figure 2, which has been regenerated.
- The reason is now given explicitly: the fixed baseline places zero probability on categories that the reference records as occurring, so its divergence is governed by the smoothing constant ε used in the calculation. Its magnitude therefore registers the presence of impossible predictions rather than the size of a behavioral error, which is why a ratio between the two columns is not a measurable effect size.
- The revised passage instead reports both magnitudes directly—below 0.03 nats for every ATUS-grounded condition against 9.92 to 11.48 for the fixed baseline—and explains why the baseline’s value is large, so the comparison rests on the measured quantities rather than on a ratio between them.

Revised text (Section 3.2):

The fixed baseline assigns deterministic activities by hour-of-day rules (e.g., sleeping 23:00–06:59, work 09:00–16:59) without demographic conditioning, which produces large divergence against the ATUS reference—especially for the retired (O2) and unemployed (O4) strata, whose schedules differ most from the employed-adult assumption. Every ATUS-grounded condition stays below 0.03 nats, while the fixed baseline ranges from 9.92 to 11.48. The gap is one of kind rather than degree: the baseline places zero probability on categories the reference records as occurring, so its divergence is governed by the smoothing constant ε and registers the presence of impossible predictions rather than the size of a behavioral error.

Reviewer #2

I found this manuscript interesting and relevant for the SoftwareX audience. The idea of combining LLM-based agents with ATUS data to simulate occupant behaviour is timely, and I appreciate the effort put into developing an open-source platform that can be integrated with different building simulation environments. The software itself appears well designed, modular and well documented. That said, I think the manuscript would benefit from a substantial revision before publication.

Comment 1. *Several sections, especially the illustrative example, the validation discussion and the impact section, are much longer than necessary. I think the manuscript could be significantly shortened without losing any important information, making the focus more clearly on the software itself.*

Response.

- This suggestion has been adopted, and the sections identified have been shortened. The walk-through loses the O4 contrast paragraph, which duplicated the Tier 2 results, and shortens the commentary following the second agent quotation; the Tier 2 discussion loses the per-stratum

narration that restated Table 7 row by row; and the architecture overview and input description have been compressed where they restated Figure 1 and Table 3.

- Together these remove roughly 300 words. Illustrative Examples, which now also carries the former Validation section, is 117 words shorter than those two sections were at submission, despite absorbing the qualifications requested elsewhere in this round.
- Impact was reworded rather than shortened (367 to 380 words), since the SoftwareX template identifies it as the main section that reviewers weight most heavily. Its “foundation” and “previously impractical” claims were removed per Comment 3, but the requested qualifications on the five-seed counts were added.
- The overall word count is addressed under Editor Comment 2.

Comment 2. *I also feel that the validation of the LLM component could be discussed more carefully. The comparison between the generated activity schedules and the ATUS data is convincing, but the validation of the reasoning process is mostly based on showing that different personas behave differently under the same conditions. While this demonstrates internal consistency, it does not necessarily demonstrate that the generated behaviour is realistic. I would suggest acknowledging this limitation more explicitly.*

Response.

- This distinction is accepted without qualification, and it is now the first of the three constraints stated on Tier 2, identified as the most fundamental.
- The text separates the two claims directly: that agents instantiated from different demographic priors behave differently, and in directions consistent with those priors, is internal consistency; behavioral realism is explicitly not established, and is identified as requiring comparison against measured occupant data.
- Establishing realism is identified as requiring comparison against measured occupant data, and is stated as future work. The abstract and Conclusions have been aligned so the same distinction is carried wherever Tier 2 is summarized.

Revised text (Section 3.3):

Three constraints bound the interpretation. First, and most fundamentally, Tier 2 establishes *internal consistency*: agents instantiated from different demographic priors behave differently, and in directions consistent with those priors. It does not establish behavioral realism, which would require comparison against measured occupant data and is left to future work.

Revised text (Section 5):

Tier 1 confirms that ATUS grounding produces empirically calibrated activity distributions; Tier 2 shows that demographic priors propagate into differentiated agent reasoning, though it does not by itself establish that the resulting behavior is realistic.

Comment 3. *Along the same lines, some of the claims throughout the manuscript are stronger than what is actually supported by the presented results. Expressions suggesting that the platform provides a new foundation for occupant modelling or significantly advances the field could be toned down. The work is certainly promising, but I think the conclusions should remain closer to what has been demonstrated.*

Response.

- The four passages carrying these claims have been revised together so the manuscript is consistent: the abstract, Section 1, Section 4, and Section 5.
- The abstract and Section 1 no longer state that the platform is “expected to advance how the human behavioral layer is modeled” or provides “a shared foundation”; both now describe a reusable, openly available implementation that other groups can evaluate and build upon. The word “foundation” has also been removed from the gap statement earlier in Section 1, which now refers to a “shared, survey-grounded starting point”, and the two universal negatives in that sentence (“none provides... none releases”) have been softened to “few”.
- In Section 4, “realistic” has been replaced by “demographically differentiated”, which is what the results support; “empirically validated behavioral baseline” has become “documented, survey-grounded baseline”; and the closing claim about shifting a specialized capability into a foundation for the entire community has been removed. The section’s opening sentence, which claimed four lines of research were “previously impractical”, now states that the platform lowers the barrier to research that is difficult to pursue with fixed-schedule models.
- The Conclusions no longer describe a “reframing” or claim “a property that has been absent from the occupant behavior modeling literature”, a statement that could not be substantiated.

Revised text (Abstract):

BUILD OCC provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer.

Revised text (Section 5):

Beyond the implementation, the platform treats occupant behavior as a structured, demographically parameterizable experimental variable rather than an uncontrolled source of uncertainty. The ATUS-derived strata, frozen schema, and plugin registry are intended to make that variable reproducible and extendable across independent research groups.

Comment 4. *Another aspect that I think deserves more attention is the computational side of the framework. Since the platform relies on repeated LLM calls, it would be useful to include at least a brief discussion about runtime, API costs, scalability and the practical implications of simulating multiple occupants. These are important considerations for researchers who may want to adopt the software.*

Response.

- A new subsection, Section 2.7 (Computational Cost and Scalability), has been added in response to this comment and to Reviewer 1’s Comment 7; the full text is quoted there.
- It states that cost and runtime scale linearly with simulated agent-days, gives the per-agent-day call count and the response token cap from which cost follows at the provider’s current price, identifies provider latency as the wall-clock bottleneck, and describes the two paths that reduce or eliminate per-call cost.
- The practical implications of simulating multiple occupants, raised specifically in this comment, are addressed in a second paragraph of the same subsection, quoted below.

Revised text (Section 2.7):

These figures are per agent. Because agents are independent—each carries its own persona, memory, and schedule, and the REST API is stateless—a multi-occupant household is simulated

by running one agent per occupant, so call volume scales linearly with household size (a four-occupant household is roughly 384 calls per simulated day) while the agents themselves can be run in parallel across processes. Interaction between occupants, such as negotiation over a shared thermostat, is not modeled in this version and is identified as future work in Section 5.

Comment 5. *Finally, although the authors mention that the framework is currently based on US demographic data, I would appreciate a slightly more detailed discussion on how easily the approach could be transferred to other countries or datasets.*

Response.

- Section 2.6 now describes transfer concretely, as a data-substitution task rather than a redesign: the scheduler consumes a table of activity probabilities by stratum, hour, and day type, so any national time-use survey following harmonized diary conventions can be substituted behind `BaseScheduler`. The Multinational Time Use Study and the European HETUS series are named as candidates.
- What does *not* transfer automatically is also stated, since it is the more useful information for an adopter: the persona layer’s appliance-ownership and income priors are drawn from a US survey (RECS) and would require a national equivalent.

Revised text (Section 2.6):

Transfer to another country is a data-substitution task rather than a redesign: the scheduler consumes a table of activity probabilities by stratum, hour, and day type, so any national time-use survey following harmonized diary conventions—the Multinational Time Use Study (Fisher et al. 2013) or the European HETUS series (Eurostat 2019), for example—can be substituted by regenerating that table behind `BaseScheduler`. What does not transfer automatically is the persona layer, whose appliance-ownership and income priors come from a US survey (RECS) and would need a national equivalent.

Overall, I think this is a valuable software contribution that fits well within the scope of SoftwareX. After addressing the points above and streamlining the manuscript, I believe it would make a strong publication.

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research

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Abstract

Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant's personal history, current context, and the type of energy signal being delivered. This study presents BUILD OCC, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset covering 16,684 respondents. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface—Python library, REST API, and Model Context Protocol server—so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers show that ATUS-grounded sampling reproduces empirically calibrated activity distributions and that demographic priors propagate into persona-consistent agent reasoning across timesteps, establishing internal consistency across strata. BUILD OCC provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer. BUILD OCC is openly released at <https://doi.org/10.5281/zenodo.21192895> under the Apache License 2.0 and installable via `pip install buildocc`.

Keywords: Artificial Intelligence Agent, Occupant Behavior Modeling,

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Required Metadata

Current code version

Table 1: Code metadata (mandatory)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0
C2	Permanent link to code/repository	https://github.com/humanbuildingsynergy/Build0cc
C3	Permanent link to reproducible capsule	https://doi.org/10.5281/zenodo.21192895
C4	Legal code license	Apache License 2.0
C5	Code versioning system	Git
C6	Software code languages, tools, and services used	Python 3.11+
C7	Compilation requirements, operating environments & dependencies	<code>pip install buildocc</code> ; requires an LLM API key (Anthropic, OpenAI, Google), or local Ollama
C8	Developer documentation	https://github.com/humanbuildingsynergy/Build0cc/blob/main/README.md
C9	Support email for questions	wooyoungjung@arizona.edu

1. Motivation and Significance

Humans are complex systems that respond to environmental, social, and economic signals in ways that are difficult to predict, introducing substantial uncertainty in building energy use and management [1, 2]. This long-standing challenge has motivated the development of occupant behavior models that can capture the stochastic and adaptive nature of human decision-making [3]. However, most existing models rely on fixed schedules (e.g., ASHRAE 90.1 reference schedules [4]) or stochastic processes that do not comprehensively capture and reason about the occupant’s personal history, current context, or

the type of energy signal being delivered [3, 2]. Consequently, these models cannot replicate the nuanced and adaptive responses that real occupants exhibit in response to various conditions.

The emergence of large language models (LLMs) and their architecture for reasoning and memory has opened new opportunities for artificial intelligence (AI) agents to simulate human-like decision-making processes. These agents are capable of contextual reasoning, memory retention, and adaptive behavior, making them suitable for modeling complex systems, including occupants in buildings. Recent studies demonstrated the potential of LLM-integrated agents to simulate occupant behavior in building energy contexts, showing that they can produce more realistic and adaptive responses to various energy signals compared to traditional models [3, 5, 6, 7, 8, 9]. Specifically, Nicholas et al. [10] proposed agents that adopt the generative agent memory architecture of Park et al. [11]—combining a persona-based core memory, a timestep-level memory stream, and periodic reflection—and demonstrate that this representation produces qualitatively different energy demand patterns than fixed-schedule or rule-based occupant models. Other recent approaches include agents conditioned on IEQ sensor readings for comfort and health feedback [12], personality-driven agents for negotiating shared thermostat setpoints [13], and a Thought-Action-Reflection-Journal framework for modeling energy behavior under dynamic pricing [14]. However, these agents are instantiated from hand-crafted profiles or small convenience samples [15]. In addition, few provide quantitative validation against empirical activity benchmarks, and few release an open-source platform for integration with building energy tools—leaving the community without a shared, survey-grounded starting point for reproducible occupant behavior research. This study aims to fill this gap by introducing BUILD OCC, an open-source platform for LLM-based occupant behavior simulation and building energy management. BUILD OCC grounds agent personas and activity schedules in the American Time Use Survey (ATUS) [16], a nationally representative dataset covering tens of thousands of diary respondents across income, employment, and household-type strata. Each agent maintains a memory stream that accumulates timestep-level observations and periodically synthesizes higher-order behavioral insights through reflection—enabling coherent, adaptive decision-making across a simulation day rather than independent per-timestep responses. Users can drive agents through building simulation timesteps using any supported LLM provider and receive structured behavioral decisions—thermostat adjustments, device control actions, and demand response reactions—that feed directly into EnergyPlus [17] co-simulation, Home Assistant automations, or custom building management systems through a three-layer interface (Python library, REST API, and

MCP server). Abstract base classes and a plugin registry allow the community to extend the platform with new occupant strata, behavioral data sources, and memory architectures. Consequently, BUILD OCC offers a reusable and extensible implementation of the occupant behavioral layer, released openly so that it can be evaluated and built upon by other research groups.

2. Software Description

Following an overview of the platform architecture, the platform is described in six parts: the ATUS behavioral grounding that drives activity scheduling and persona sampling; the inputs required at simulation initialization; the per-timestep LLM reasoning pipeline; the demand response signal interface; the extensibility mechanisms that allow the platform to be adapted to new research contexts; and the computational cost of running it.

2.1. Architecture Overview

BUILD OCC is organized around two levels of architecture: an internal *agent component architecture* that defines how occupant agents perceive, reason, and act within a simulation timestep, and a *platform interface* that exposes this functionality to external building energy tools (Figure 1).

2.1.1. Agent Component Architecture

The BUILD OCC occupant agent is composed of six interacting components:

- **Persona** — a structured demographic profile assigned to one of four strata (O1–O4; Table 2), with appliance ownership, income bracket, and thermal comfort tolerance sampled from RECS [18] probability distributions.
- **ActivityScheduler** — samples the occupant’s activity at each 15-minute timestep from stratum- and hour-specific probability distributions derived from ATUS 2022–2023 diary records (16,684 respondents), with weekday/weekend stratification (Appendix A).
- **MemoryStream** — an accumulating log of observations, actions, and reflections ranked by recency and LLM-assigned importance following Park et al. [11]; top- k entries are injected into each timestep prompt and periodically synthesized into higher-order behavioral insights through a reflection call (Appendix B).
- **LLM Reasoning Engine** — assembles a prompt from the persona description, current ATUS activity, environment state, and retrieved

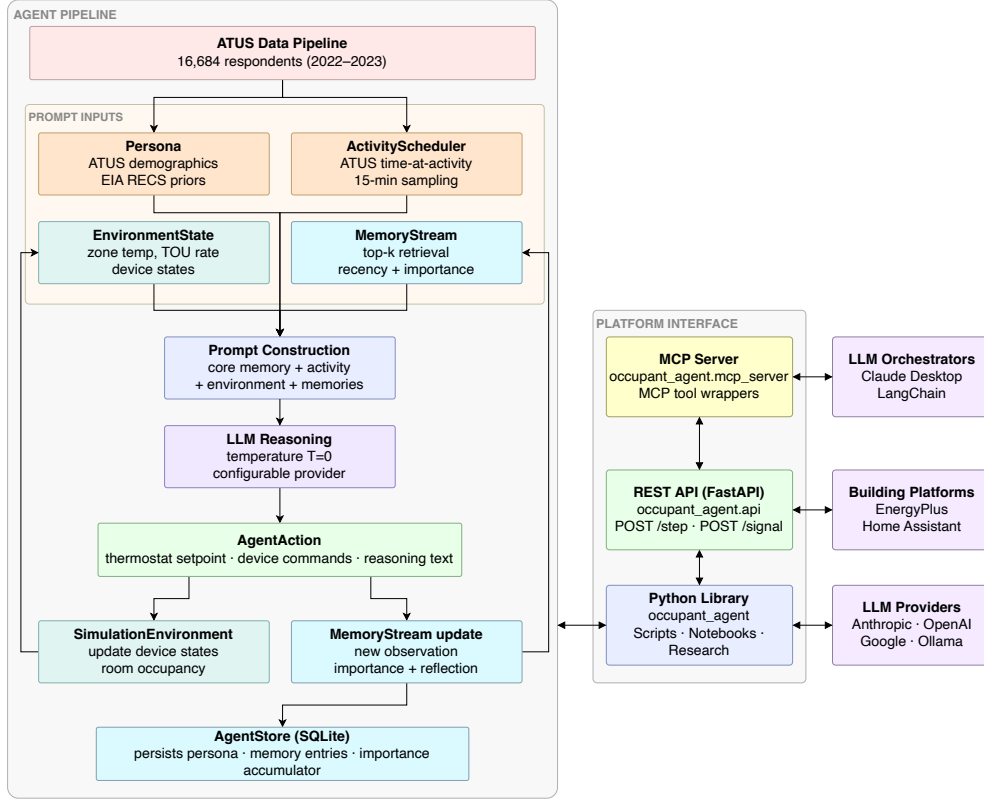


Figure 1: Architecture of BUILD OCC. *Left*: per-timestep agent pipeline showing how ATUS-grounded activity codes, retrieved memories, and environment state are assembled into an LLM prompt and how the resulting **AgentAction** updates both the simulation environment and the memory stream. *Right*: three-layer platform interface through which external building energy tools and LLM orchestration frameworks access the agent.

memories, then queries a configurable LLM provider (Anthropic, OpenAI, Google, or local Ollama) at temperature $T=0$ for reproducibility, returning a structured action with a natural-language reasoning string.

- **SimulationEnvironment** — maintains device states, room occupancy, and thermostat setpoint between timesteps: at the start of each timestep it exposes the current state to the LLM Reasoning Engine, and at the end it applies the agent’s returned action.
- **AgentStore** — persists the agent’s cognitive state (persona, memory entries, and the importance accumulator) to SQLite via SQLAlchemy after each operation. Environment state is owned by the caller and

passed fresh with each request, enabling a stateless REST API pattern that supports multi-process deployment.

2.1.2. Platform Interface

BUILD OCC is distributed as the package `buildocc` and imported as `occupant_agent`. It exposes the agent through three integration layers, each targeting a different class of integrator:

1. **Python library** (`occupant_agent`) — in-process access to all agent components with no server overhead; the recommended path for batch simulations and notebook experiments.
2. **REST API** (`occupant_agent.api`) — a stateless FastAPI [19] server exposing agent initialization, per-timestep decisions (`POST /step`), signal delivery (`POST /signal`), and state query (`GET /state`); suitable for EnergyPlus co-simulation callbacks and any platform that can issue HTTP requests.
3. **MCP server** (`occupant_agent.mcp_server`) — wraps the REST API as Model Context Protocol [20] tools, so any MCP-compatible host (Claude Desktop, LangChain, Home Assistant) can drive a simulation without additional integration code.

2.2. ATUS Behavioral Grounding

Activity schedules are derived from ATUS 2022–2023 microdata (16,684 respondents). Time-at-activity probabilities are aggregated across eight categories (sleeping, work, food preparation, laundry, television viewing, eating, exercise, and other) at 15-minute resolution, stratified by demographic group and day type (weekday / weekend). The method follows the “what is the person doing at this moment?” framing: for each diary episode $[t_{\text{start}}, t_{\text{stop}}]$, the method queries whether the episode covers the half-hour mark of each hour h , then computes a weighted probability across respondents using ATUS person-level weights. ATUS records a single primary activity per episode, so simultaneous secondary activities—watching television while preparing food, for example—are not represented in the scheduler’s output.

Four demographic strata are defined (Table 2), selected to span the variation in building occupancy and demand response participation that is most relevant to residential energy research (Appendix C).

An important methodological choice is the *in-progress* framing: rather than sampling from the start-time distribution (which biases toward short activities), the method evaluates which activity is in progress at the half-hour mark of each hour, reflecting the actual time spent in each activity state

Table 2: Demographic strata

ID	Label	ATUS filter
O1	Employed single adult	Full-time employed, lives alone, age 25–44
O2	Retired couple	Age 65+, not employed
O3	Employed parent	Full-time employed, children present, age 35–54
O4	Unemployed adult	Not employed, age 25–44

rather than the frequency of starts. The formal probability computation is described in Appendix A.

Work-from-home (WFH) behavior is handled separately: a per-agent WFH probability encodes the fraction of working days spent at home (drawn from ATUS 2022–2023 telework questions), and a per-day Bernoulli draw determines WFH status for each simulated day. The WFH flag is included in the agent’s per-timestep prompt so that the LLM reflects both the ATUS activity code and the work location, preventing the scheduling model from assigning “work” activities that place the agent outside the home on days it is working remotely.

Each stratum uses stratum-specific appliance ownership priors from EIA RECS [18, 21] and an income-bracket-modulated comfort tolerance (in °C), reflecting the documented tendency of lower-income households to accept wider thermal deviations to reduce HVAC costs [22, 18, 23, 24]. Income brackets are currently sampled uniformly within each stratum’s RECS income range.

2.3. User-Provided Inputs

Table 3 lists all inputs required at initialization; the only per-timestep value the calling application must supply is zone temperature, since it depends on the building’s physical response to occupant actions. To lower the barrier to first use, the library ships preset helpers for a representative household device inventory and room layout. A command-line interface also ships preset helpers and a YAML-driven configuration path for users who prefer not to write Python.

Users without a live co-simulation link can supply a constant zone temperature (suitable where thermal dynamics are not the focus) or a pre-computed CSV schedule indexed by datetime; a sample file covering one week of EnergyPlus output is included in the repository.

Table 3: User-provided inputs to BUILD OCC. All are specified once at initialization, except zone temperature, which the calling platform supplies at each timestep.

Input	Type	Description
Stratum	String	Demographic profile: O1–O4
Seed	Integer	Controls persona attribute sampling, activity scheduling, and work-from-home day draws; the same seed produces an identical agent across runs.
LLM provider	String	Anthropic, OpenAI, Google, or Ollama
Device inventory	List	Appliance IDs, initial on/off states, rated power (W)
Room list	List	Rooms the occupant can occupy
Thermostat setpoint	Float	Initial comfort setpoint (°C)
Zone temperature	Source	Fixed value, CSV file, or live building platform
Outdoor temperature	Source	Fixed value, CSV file, EPW file, or callable
TOU rate schedule	Source	Fixed value, CSV file, or tariff function

2.4. LLM Reasoning Pipeline

With all inputs initialized, at each 15-minute simulation timestep the agent executes the following five-step pipeline:

1. **Activity resolution:** the current ATUS code is translated to a natural-language description and an occupancy flag using a lookup table of ATUS activity codes.
2. **Memory retrieval:** up to five entries are selected by a retrieval score that combines recency (exponential decay, half-life 24 h) and a 1–10 importance score the LLM assigns to each entry at the time of recording — ensuring that consequential events remain retrievable even as they age (see Appendix B), following Park et al. [11].
3. **Prompt construction:** a system prompt encodes the persona’s stable identity — demographics, appliance ownership, and comfort tolerance. A user prompt encodes the current environment state, ATUS activity, and the retrieved memories.
4. **LLM call:** the prompt is sent to a configurable provider (Anthropic Claude, OpenAI GPT, Google Gemini, or Ollama) at temperature $T=0$ for reproducibility. The model selects from four action types (Table 4), returning a JSON object with target, value, and a one-sentence memory note.

5. **Memory update:** the observation and its LLM-assigned importance score are appended to the stream. If the cumulative accumulator exceeds a threshold (default 100), a reflection pass is triggered in which a higher-capability model distills recent observations into three higher-order insights stored as reflection entries.

Table 4: The four action types returned by `step()` at each timestep, with their decision trigger and a representative example. Demand response is handled separately via `receive_signal()` (Section 2.5).

Action type	Trigger	Example
<code>do_nothing</code>	No environmental or activity condition warrants intervention	Agent is sleeping; no device is running
<code>adjust_thermostat</code>	Zone temperature deviates from comfort tolerance, or TOU rate context	Raises setpoint during peak window to reduce HVAC cost
<code>toggle_device</code>	Activity–appliance alignment (on) or inactivity (off)	Turns off lights when leaving for work
<code>move_room</code>	Activity context suggests transitioning to a different room	Moves to bedroom at 23:00 for sleeping

Setting the sampling temperature to $T=0$ makes the model return its single most probable output rather than sampling among alternatives, so a fixed provider, model version, and prompt reproduce the same action on every run. This determinism does not extend across providers, or across successive versions of a hosted model, because it is relative to one specific set of model weights. Reproducible studies therefore record the provider and model version alongside the seed, and a locally hosted open-weight model can be pinned where results must remain reproducible over time.

The pipeline above governs autonomous per-timestep behavior; demand response scenarios additionally require the ability to deliver external signals to the agent, which is handled by the interface described in the following subsection.

2.5. Demand Response Signal Interface

Alongside the per-timestep pipeline, BUILD OCC provides a dedicated signal delivery interface that can be invoked at any point in a simulation. Three signal types are supported (Table 5), following established behavioral intervention typologies [22, 25, 26]. Signals are delivered by passing a signal type,

a free-text message, and the current environment state to the agent, which returns a response code (*accepted*, *rejected*, or *deferred*) alongside a natural-language reasoning string drawn from the agent’s memory context. Keeping the three types mutually exclusive enables factorial experimental designs in which signal type and demographic stratum are the primary independent variables.

Table 5: Demand response signal types supported by BUILD OCC, classified by behavioral lever and illustrated with representative example messages. All three types share the same delivery mechanism and response codes (*accepted*, *rejected*, *deferred*).

Type	Behavioral lever	Example message
A — Direct command	Regulatory compliance	“Turn off your HVAC for 30 min to support grid stability during today’s peak demand event.”
B — Educational	Economic information	“Your HVAC costs 3× more per kWh at the current peak rate — raising the setpoint 1°C saves an estimated \$0.35 today.”
C — Social norm	Peer comparison	“75% of similar households in your area have reduced HVAC use during today’s peak event.”

The extensibility mechanisms that allow all of these components to be adapted for new research contexts are described in the following subsection.

2.6. Platform Extensibility

BUILD OCC is designed for adaptation across research contexts without modifying the core library. Extensibility is organized along three independent axes:

- **Occupant population** (`BasePersona`): new demographic profiles — commercial occupants, hotel guests, non-US populations, or any custom stratum — are registered by implementing `BasePersona`.
- **Activity grounding** (`BaseScheduler`): the built-in scheduler uses ATUS microdata; any other grounding source (a Markov chain, a physics-based occupancy model, or alternative activity data) can be substituted by implementing `BaseScheduler`.

- **Memory architecture** (`BaseMemoryStream`): the built-in recency-plus-importance retrieval can be replaced with embedding-based retrieval or any other scheme by subclassing `BaseMemoryStream`.

Extensions are registered via Python’s package entry-point mechanism and discovered automatically at import time, allowing third-party packages to extend the platform without forking the repository:

Listing 1: `pyproject.toml` entry-point declarations for a third-party stratum and scheduler.

```
[project.entry-points."occupant_agent.strata"]
P5 = "my_package.personas:LowIncomeElderlyAlone"

[project.entry-points."occupant_agent.schedulers"]
homer = "my_package.schedulers:HomerScheduler"
```

Transfer to another country is a data-substitution task rather than a re-design: the scheduler consumes a table of activity probabilities by stratum, hour, and day type, so any national time-use survey following harmonized diary conventions—the Multinational Time Use Study [27] or the European HETUS series [28], for example—can be substituted by regenerating that table behind `BaseScheduler`. What does not transfer automatically is the persona layer, whose appliance-ownership and income priors come from a US survey (RECS) and would need a national equivalent.

Researchers can inject additional context into the LLM reasoning pipeline — such as predicted mean vote (PMV) comfort indices [29], language-based comfort and heat-strain signals [30, 31], or indoor air quality readings — either by extending the agent class or by passing a free-text string at each timestep; both paths are available through the REST API and MCP server. The environment state and signal schemas are fixed to ensure reproducibility, and a test utilities module provides mock implementations and conformance tests for extension authors.

2.7. Computational Cost and Scalability

Because the agent queries a language model at nearly every timestep, cost and runtime scale linearly with simulated agent-days. A 15-minute timestep gives up to 96 calls per simulated day per agent, plus occasional reflection calls, with responses capped at 512 tokens; the monetary cost therefore follows directly from the provider’s per-token price at the time of use, and wall-clock time is dominated by provider latency rather than local computation. Two mechanisms reduce this. Any local open-weight model served through Olama removes per-call cost entirely, at the price of reduced reasoning quality,

and scheduler-only studies—including the Tier 1 validation reported in Section 3.2—require no model calls at all. Cost, rather than platform capability, is what bounded the simulation lengths reported in Section 3.

These figures are per agent. Because agents are independent—each carries its own persona, memory, and schedule, and the REST API is stateless—a multi-occupant household is simulated by running one agent per occupant, so call volume scales linearly with household size (a four-occupant household is roughly 384 calls per simulated day) while the agents themselves can be run in parallel across processes. Interaction between occupants, such as negotiation over a shared thermostat, is not modeled in this version and is identified as future work in Section 5.

3. Illustrative Examples

This section demonstrates the platform in three parts: a walkthrough of a single agent through a demand response event, followed by two quantitative tiers. Tier 1 validates the *ActivityScheduler*, testing whether ATUS-grounded sampling produces category distributions matching empirical ATUS reference data relative to a deterministic rule-based baseline; Tier 2 validates the *LLM agent decisions*, testing whether each stratum’s demographic priors propagate into distinguishable behavioral patterns under identical environment conditions. Both are necessary because a well-grounded scheduler and a well-calibrated reasoning agent are independent failure modes — one can be correct while the other is not.

3.1. Demand response walkthrough

To illustrate agent behavior during a residential demand response event, this subsection traces an O1 agent (28-year-old employed male, income bracket 9, comfort band $\pm 1.1^\circ\text{C}$) through the peak electricity period of a summer weekday evening (18:00–19:45, TOU rate \$0.22/kWh). Zone temperatures come from pre-computed EnergyPlus output for the DOE PNNL residential prototype building [32] (single-family, Climate Zone 2B, variable-speed heat pump) included in the repository. The simulation loop is:

Listing 2: Running the BuildOcc simulation loop with the sample zone temperature CSV.

```
1 # Full imports available in examples/simulation_loop.py (  
2 repository)  
3 from datetime import datetime, timedelta  
4 agent = OccupantAgent.from_stratum("O1", seed=42)  
5 scheduler = ActivityScheduler(stratum="O1", seed=42)
```

```

1
2
3
4
5     zone_temp = zone_temp_from_csv(...)    # pre-computed
6         EnergyPlus zone temperatures
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```

At 18:00 (Step 1), the CSV-sourced zone temperature is 23.9°C and the thermostat is set to 22°C. With outdoor temperatures exceeding 38°C, the HVAC is running continuously to hold the setpoint. The agent is engaged in computer leisure and proactively raises the thermostat setpoint to 24°C to reduce peak-rate HVAC run time, reasoning:

“The indoor temp is 23.9° C and the thermostat is set to 22° C — the HVAC is running hard to maintain the setpoint with 39° C outside during peak rate hours (\$0.22/kWh). I’m comfortable at 23.9° C, well within my 1.1° C tolerance, so raising the setpoint to 24° C will meaningfully cut cooling costs during this expensive period.”

This response demonstrates that the persona’s income bracket and comfort tolerance surface in actionable decision-making without explicit rule encoding.

At 18:30 (Step 3), a Type B educational signal is delivered: *“Your HVAC is running during peak hours (4–9pm) at \$0.22/kWh—three times the off-peak rate. Raising the setpoint by 1° C saves ~\$0.35 today.”* The agent accepts the signal but references its own prior action stored in memory:

“I already raised the thermostat to 24° C earlier today and it’s working—I’m at 23.9° C and comfortable within my tolerance. The math is clear (\$0.35 savings) and the effort is zero since I’ve already taken this action.”

This cross-timestep coherence—the agent recalling and acting on its own prior decision—is what a fixed schedule cannot represent.

Over the remainder of the evening the agent maintains the raised setpoint without further intervention; importance ratings for routine observations stay low (2–3 out of 10).

3.2. Tier 1: Activity Schedule Fidelity

The ATUS-grounded activity scheduler is validated against a deterministic rule-based baseline using KL divergence between simulated activity distributions and ATUS empirical distributions. For each stratum, 180 days of 96 timesteps each (17,280 observations per stratum) are simulated, the empirical category distribution at each hour is computed, and compared to the ATUS reference using:

$$D_{\text{KL}}(P\|Q) = \sum_c P(c) \log \frac{P(c)}{Q(c) + \varepsilon} \quad (1)$$

where P is the ATUS reference distribution, Q is the simulated distribution, and $\varepsilon = 10^{-9}$ is a smoothing constant. Table 6 and Figure 2 report the mean per-hour KL divergence averaged across all 24 hours and both day types.

Table 6: Mean per-hour KL divergence (nats): ATUS-grounded vs. rule-based scheduler. Computed over 180 simulated days, 96 timesteps each, weekday/weekend split. Lower is better.

Stratum	ATUS scheduler	Fixed baseline
O1 (Employed single adult)	0.0181	10.11
O2 (Retired couple)	0.0253	11.48
O3 (Employed parent)	0.0184	9.92
O4 (Unemployed adult)	0.0092	11.27

The fixed baseline assigns deterministic activities by hour-of-day rules (e.g., sleeping 23:00–06:59, work 09:00–16:59) without demographic conditioning, which produces large divergence against the ATUS reference—especially for the retired (O2) and unemployed (O4) strata, whose schedules differ most from the employed-adult assumption. Every ATUS-grounded condition stays below 0.03 nats, while the fixed baseline ranges from 9.92 to 11.48. The gap

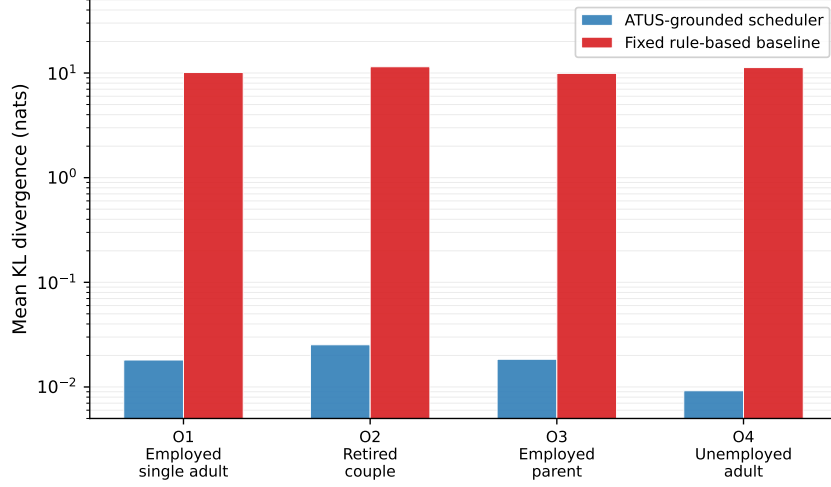


Figure 2: Mean per-hour KL divergence (nats, log scale) between simulated and ATUS empirical activity distributions over 180 simulated days. The baseline’s magnitude is governed by the smoothing constant, since it places zero probability on categories the reference records as occurring.

is one of kind rather than degree: the baseline places zero probability on categories the reference records as occurring, so its divergence is governed by the smoothing constant ϵ and registers the presence of impossible predictions rather than the size of a behavioral error.

Figure 3 illustrates what this divergence looks like category by category for stratum O1 on weekdays: the ATUS-grounded simulation (panel a) closely tracks the reference distributions, while the fixed rule-based baseline (panel b) diverges markedly—most notably collapsing daytime activity into sharp step functions and failing to reproduce the gradual evening wind-down.

3.3. Tier 2: Persona-Driven Behavioral Differentiation

Tier 2 tests whether the demographic priors encoded in each stratum’s persona produce distinguishable behavioral patterns under identical environment conditions, so that any observed differences are attributable to persona variation rather than environment variation. Each of O1–O4 was run for three independent seeds over one weekday (96 timesteps at 15-minute resolution) under identical environment conditions—the same EnergyPlus zone temperature, device configuration (HVAC, three lighting zones, TV, washer, dishwasher), and time-of-use (TOU) tariff (\$0.22/kWh peak, \$0.08/kWh off-peak).

Table 7 reports the resulting action-type distributions and mean thermostat setpoints during the TOU peak window (4–9 pm). O2 (retired couple) is

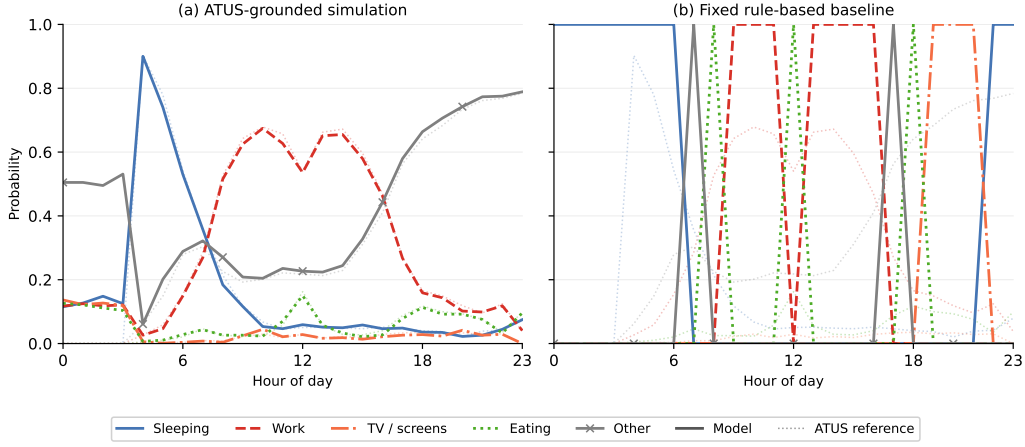


Figure 3: Hourly activity probability distributions for stratum O1 (employed single adult, weekday) across five activity categories. (a) ATUS-grounded simulation (180 simulated days): bold lines closely track the reference. (b) Fixed rule-based baseline: bold lines diverge sharply, with the baseline collapsing smooth activity transitions into unrealistic step functions and misassigning the O1 late-evening activity peak.

the most active stratum—lowest inactivity (67.4%), highest device-toggling (8.0%) and room-movement (21.5%) rates, and the highest mean peak set-point (24.3°C)—consistent with full-day home occupancy. O4 (unemployed adult; ATUS $n=107$, Appendix C) matches O2’s toggling rate but records the lowest thermostat-adjustment rate (1.4%) and peak setpoint (23.1°C). Because O4’s probability tables derive from only 107 ATUS respondents—the smallest of the four strata and the noisiest at 15-minute resolution—this pattern should be read as provisional; it may reflect sampling noise in the grounding data as much as a demographic tendency. O3 (employed parent) shows the lowest device-toggling rate (3.5%), its peak-hour schedule dominated by ATUS child-care codes, while O1 sustains a moderate profile throughout. Figure 4 visualizes these distributions.

To assess demand response participation, three standardized signals were delivered to each stratum (five seeds each) after a 2.5-hour context period at peak-rate conditions (10 simulated timesteps beginning at 16:00, giving each agent an experiential baseline in memory before signal delivery): Type A (direct command: “Turn off your HVAC for 30 minutes”), Type B (educational price signal: cost comparison at three times the off-peak rate), and Type C (social norm: “75% of similar households have reduced HVAC use”). Table 8 reports the number of accepting agents per stratum and signal type. Direct commands (Type A) drew the most acceptance for O2 (retired couple, 2 of 5

Table 7: Action-type distribution (%) per stratum: mean over three seeds, one weekday each (96 steps), giving 288 timesteps per stratum. At this sample size one percentage point corresponds to roughly three timesteps, so differences of a few tenths should not be read as meaningful. Peak SP is the mean thermostat setpoint during TOU peak hours (4–9 pm, \$0.22/kWh).

Stratum	Inactive	Thermostat	Toggle	Move	Peak SP (°C)
O1 (Employed single)	76.0	2.8	4.2	17.0	23.5
O2 (Retired couple)	67.4	3.1	8.0	21.5	24.3
O3 (Employed parent)	76.4	3.1	3.5	17.0	23.6
O4 (Unemployed)	72.9	1.4	8.0	17.7	23.1

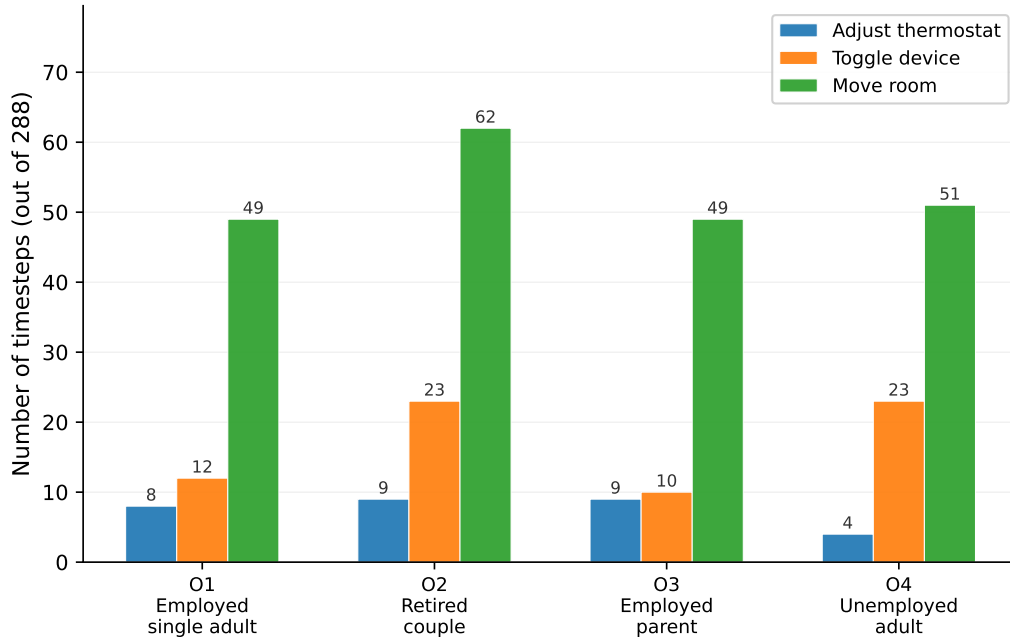


Figure 4: Active action counts per stratum across 288 timesteps (3 seeds $\times$ 96 steps, identical environment conditions). Do-nothing accounts for 67–76% of all timesteps across strata and is excluded to isolate cross-stratum differences in active decision-making. Bar labels show absolute counts.

seeds), with O1 (employed single) and O4 (unemployed adult) accepting in 1 of 5; educational price signals (Type B) were accepted by O2 and O4 in 2 of 5 seeds each, a pattern consistent with cost-aware strata responding to both authority and economic framing. O3 (employed parent) rejected all signal types across all five seeds: signal delivery at 18:30 coincides with child-care activity codes in the ATUS schedule, making that outcome an expected con-

sequence of schedule grounding rather than a general insensitivity to demand response. Social norm signals (Type C) drew no acceptance in any stratum, consistent with field evidence on the limited efficacy of peer-comparison messaging [22, 25]. With five runs per cell these counts indicate direction only; no significance testing is performed on them.

Table 8: Signal compliance per stratum and signal type, reported as the number of accepting agents out of five random seeds. Percentages are not given, since five runs per cell cannot support them. Type A: direct command; Type B: educational price signal; Type C: social norm nudge.

Stratum	Type A	Type B	Type C
O1 (Employed single)	1/5	0/5	0/5
O2 (Retired couple)	2/5	2/5	0/5
O3 (Employed parent)	0/5	0/5	0/5
O4 (Unemployed)	1/5	2/5	0/5

These results show that the ATUS demographic priors propagate into qualitatively distinct LLM-driven behavioral patterns: employment status shapes thermostat adjustment frequency, household composition modulates signal compliance, and full-day home occupancy drives device-toggling behavior. Three constraints bound the interpretation. First, and most fundamentally, Tier 2 establishes *internal consistency*: agents instantiated from different demographic priors behave differently, and in directions consistent with those priors. It does not establish behavioral realism, which would require comparison against measured occupant data and is left to future work [33]. Second, each stratum is run for only one weekday per seed (three seeds for action distributions, five for signals), which is sufficient to show directional patterns but not to establish statistical significance; multi-day designs are feasible but were constrained by API cost. Third, Tier 2 uses a single environment configuration (fixed weather, tariff structure, and household composition), which limits generalizability across different climate zones and occupancy contexts. The behavioral differences across strata also reflect a structural property of the platform: both the ATUS-grounded activity schedule and the LLM persona reasoning are conditioned on the same demographic stratum, a coupling that ensures internal consistency between the scheduling and decision-making components.

4. Impact

BUILD OCC lowers the barrier to four lines of research that are difficult to pursue with fixed-schedule occupant models:

Survey-grounded behavioral agents. BUILD OCC packages population-calibrated occupant agents as a reusable component rather than a per-project infrastructure investment. Research groups can introduce demographically differentiated occupant behavior into any building energy study without constructing a grounding pipeline from scratch. The ATUS-derived activity distributions and RECS-conditioned personas provide a documented, survey-grounded baseline that any researcher can adopt, extend, or cite as a reference condition.

Demographic behavioral experiments. BUILD OCC’s combination of ATUS-grounded demographic strata, persona income brackets, and configurable intervention signals enables controlled factorial studies of occupant behavior. Unlike fixed-schedule models, each agent reasons from its demographic persona and accumulated memory at every timestep, producing adaptive responses that reflect personal history and current context rather than predetermined patterns. Because both the activity schedule and the agent’s reasoning are conditioned on the same demographic persona, researchers can hold environment and intervention constant while varying stratum — or vary the intervention while holding stratum constant — isolating the contribution of socioeconomic and life-stage factors to any behavioral outcome. The Tier 2 signal experiment (Table 8) illustrates this: Type B educational signals drew equal or higher acceptance than direct commands for cost-sensitive strata (O2 and O4: 2 of 5 seeds each), while O1 (employed single adult) responded only to Type A direct commands (1 of 5) and O3 (employed parent) rejected all signal types — differences in direction that a fixed schedule cannot produce at all, having no mechanism to respond to a signal, though with five seeds per cell they are indicative rather than precise. The signal types correspond to established intervention typologies in behavioral energy research [22, 25], making results directly comparable to field studies.

Platform-neutral integration. The MCP server layer means that any tool with MCP support—including Claude Desktop, LangChain, energy scheduling systems [34], and custom orchestration frameworks—can drive a building simulation with demographically differentiated occupant behavior without writing integration code. The REST API provides an equivalent path for EnergyPlus co-simulation and Home Assistant integrations [5, 8].

Community extensibility. The plugin registry and abstract base classes lower the barrier for researchers who want to add custom demographic strata, alternative behavioral data sources or activity grounding pipelines, or novel memory architectures. The conformance test suite enables rapid validation of extensions before publication.

5. Conclusions

BUILD OCC delivers a survey-grounded, three-layer occupant agent platform that the building energy community can use, extend, and build upon without reconstructing the behavioral layer for each study. Tier 1 confirms that ATUS grounding produces empirically calibrated activity distributions; Tier 2 shows that demographic priors propagate into differentiated agent reasoning, though it does not by itself establish that the resulting behavior is realistic.

Beyond the implementation, the platform treats occupant behavior as a structured, demographically parameterizable experimental variable rather than an uncontrolled source of uncertainty. The ATUS-derived strata, frozen schema, and plugin registry are intended to make that variable reproducible and extendable across independent research groups.

Current limitations include a single-action constraint per timestep; ATUS-grounded strata covering US demographic profiles only; activity grounding restricted to primary activities, since ATUS records only one activity at a time; and a memory-importance score that the agent assigns to itself and then reuses for retrieval, without external calibration or a feedback path that would correct a mis-scored entry. Subsequent phases extend the platform with multi-agent household simulation, non-US occupancy data, and external behavioral benchmarks including ecobee Smart Thermostat field data [35] matched to the four ATUS strata.

CRedit Author Statement

Wooyoung Jung: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares that there are no competing interests.

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A. ATUS Activity Probability Computation

This appendix provides the formal description of the activity probability computation summarized in Section 2.

Episode-overlap framing. ATUS diaries record each respondent’s day as a sequence of episodes, each with a start time t_s (minutes since midnight), stop time t_e , and a six-digit activity code in ATUS. For a 15-minute simulation slot centered at time τ , the scheduler identifies which activity episode was in progress at the midpoint $\tau + 7.5$ min:

$$c(r, \tau) = \text{activity code of respondent } r \text{ such that } t_s \leq \tau + 7.5 \text{ min} < t_e \quad (\text{A.1})$$

where r indexes individual diary respondents. This in-progress query avoids the start-time bias that would overrepresent short activities (e.g., a 5-minute meal counts only if the midpoint falls within it).

Category probability. Eight activity categories are defined (Table A.1). The probability of category k at hour h , for demographic stratum s and day type $d \in \{\text{weekday, weekend}\}$, is:

Table A.1: The eight broad activity categories used in the two-step ATUS sampling procedure, with their ATUS code filter, the number of Tier-3 codes mapped to each category, and representative subcategories.

Category	ATUS code filter	Codes	Representative subcategories
Sleeping	prefix 0101	3	Sleeping; Sleeplessness
Work	prefix 05	8	Work (main job); Work (other job); Socializing at job
Food preparation	prefix 0201	4	Food/drink preparation; Food presentation; Kitchen clean-up
Laundry	code 020202 exactly	1	Laundry
Television	code 120301 exactly	1	Watching TV
Eating	prefix 1101	2	Eating and drinking
Exercise	prefix 13	29	Aerobics; Basketball; Biking; Swimming; Weightlifting
Other	residual	172	Grooming; Reading; Computer use; Social communication; Travel

$$P(k \mid s, d, h) = \frac{\sum_{r \in R(s, d)} w_r \cdot \mathbf{1}[c(r, h:30) \in k]}{\sum_{r \in R(s, d)} w_r} \quad (\text{A.2})$$

where $R(s, d)$ is the set of diary respondents matching stratum s on day type d ; w_r is the ATUS person-level survey weight (provided by BLS to correct for sampling design); $\mathbf{1}[\cdot]$ is the indicator function that equals 1 when respondent r was engaged in a category- k activity at the half-hour mark $h:30$ of hour h ; and the denominator normalizes the weighted counts to probabilities. Evaluating Equation (A.2) for all $h \in \{0, \dots, 23\}$ and all 8 categories yields a 24×8 weighted probability table per stratum per day type, which is bundled with the installed package.

Code-level sampling. Once a category k is sampled from $P(k \mid s, d, h)$, a specific activity code is drawn from within that category proportional to the total diary minutes logged across all stratum respondents for each code. This gives common activities realistic relative frequencies (e.g., sleeping in a bed is drawn far more often than napping on a sofa, even though both map to the “sleeping” category).

Work-from-home probability. Finally, the WFH probability p_{WFH} for each stratum is estimated from the ATUS `TEWHERE` variable — the “where were you when you did this activity?” field — restricted to employed respondents who worked on their diary day. p_{WFH} equals the survey-weighted fraction of those respondents who reported working at home for any work episode, computed separately for each stratum. A per-day Bernoulli draw from p_{WFH} then determines whether the simulated agent works from home on a given day.

B. Memory Retrieval and Reflection Mechanics

This appendix provides the mathematical detail for the memory retrieval and reflection logic implemented in `MemoryStream`, following the generative agent architecture of Park et al. [11].

Retrieval scoring. At each simulation timestep t , every stored memory entry m receives a combined retrieval score $s(m, t) \in [0, 1]$ that balances how recent the entry is against how important it was judged to be when written:

$$s(m, t) = 0.5 \tilde{r}(m, t) + 0.5 \tilde{i}(m) \quad (\text{B.1})$$

where $\tilde{r}(m, t)$ is the normalized recency score and $\tilde{i}(m)$ is the normalized importance score, both in $[0, 1]$ and weighted equally.

The recency component decays exponentially with a 24-hour half-life:

$$\tilde{r}(m, t) = 2^{-(t-t_m)/24\text{ h}} \quad (\text{B.2})$$

where t_m is the simulation time (in hours) at which entry m was written, and $(t - t_m)$ is the elapsed time in hours since writing. The score equals 1.0 at the moment of creation and halves every 24 simulation hours, so memories from the previous day retain roughly half their recency weight while week-old entries contribute near zero.

The importance component normalizes the integer score $i(m)$ assigned by the LLM at write time:

$$\tilde{i}(m) = \frac{i(m)}{10} \quad (\text{B.3})$$

where $i(m) \in \{1, \dots, 10\}$ is an integer on a scale from 1 (routine, nothing unusual) to 10 (rare, extreme situation), self-assigned by the agent’s LLM in the same JSON response that produces the behavioral action. The top- k entries ranked by $s(m, t)$ are injected into the next prompt ($k = 5$ by default).

Reflection trigger. An importance accumulator A tracks the cumulative importance of all entries written since the last reflection. After each entry m is stored, $A \leftarrow A + i(m)$. When $A \geq \theta$ (default $\theta = 100$), reflection is triggered: a dedicated LLM call receives the 30 most recent entries by simulation time and returns exactly three synthesized behavioral insights (e.g., “this occupant pre-cools before peak hours to avoid peak tariffs”). Each insight is written back into the stream as a new entry with importance 9.0, and A resets to zero. At average importance 7–10 (high-salience events), reflection fires roughly every 10–14 timesteps (≈ 2.5 – 3.5 simulation hours); at average importance 2–3 (routine observations), every 33–50 timesteps (≈ 8 – 12 simulation hours).

C. Demographic Strata Selection Rationale

The four strata were selected to maximize variation in the two behavioral dimensions most consequential for residential demand response research: *peak-hour occupancy schedule* and *household composition*. The selection is grounded in the ATUS 2022–2023 microdata that drives the platform itself.

Table C.1 summarises the occupancy metrics and sample sizes derived from ATUS 2022–2023 ($n = 16,684$ respondents). O1 and O3 are absent from the home for the majority of peak grid hours (09:00–17:00), while O2 and O4 are home throughout the day: 59.8% and 61.4% of O1 and O3 weekday hours in

that window are spent in work activity, against 0.4% and 7.9% for O2 and O4. Within each occupancy cluster, household composition is varied to capture differences in thermostat authority and social dynamics: single-person (O1, O4), two-adult couple (O2), and family with children (O3) [22, 25]. O4 has the smallest sample ($n = 107$); its 15-minute probability tables are the noisiest of the four strata. Additional strata (multi-adult non-family, student dormitories) would draw from the remaining 10,643 “other” respondents but would fragment already thin subsamples without a clearly defined DR participation profile.

Table C.1: ATUS 2022–2023 occupancy metrics and sample sizes by stratum. % working 09–17 is the average weighted probability of being engaged in work activity during peak hours on weekdays (Appendix A). Work location columns are derived from the ATUS TEWHERE variable restricted to employed respondents on diary days with a work episode.

ID	Stratum	% working 09–17 (wkdy)	% at workplace	% at home	n
O1	Employed single adult	59.8	70.2	24.3	1,393
O2	Retired couple	0.4	8.1	50.1	2,351
O3	Employed parent	61.4	68.9	25.7	2,190
O4	Unemployed adult	7.9	0.0	98.4	107

Current executable software version

Table 9: Software metadata (mandatory)

Nr.	Software metadata description	Please fill in this column
S1	Current software version	v1.0.0
S2	Permanent link to executables of this version	https://pypi.org/project/buildocc/
S3	Legal software license	Apache License 2.0
S4	Computing platforms / OS	Linux, macOS, Windows (Python 3.11+)
S5	Installation requirements & dependencies	pip install buildocc; LLM provider API key required
S6	Link to user manual	https://github.com/humanbuildingsynergy/Build0cc/blob/main/README.md
S7	Support email for questions	wooyoungjung@arizona.edu

References

- [1] T. Hong, S. C. Taylor-Lange, S. D'Oca, D. Yan, S. P. Corgnati, Advances in research and applications of energy-related occupant behavior in buildings, *Energy and Buildings* 116 (2016) 694–702. doi:10.1016/j.enbuild.2015.11.052.
- [2] D. Yan, W. O'Brien, T. Hong, X. Feng, H. B. Gunay, F. Tahmasebi, A. Mahdavi, Occupant behavior modeling for building performance simulation: Current state and future challenges, *Energy and Buildings* 107 (2015) 264–278. doi:10.1016/j.enbuild.2015.08.032.
- [3] M. Arslan, S. Munawar, Large language models in building energy applications: A survey, *Energy & Buildings* 352 (2026) 116800. doi:10.1016/j.enbuild.2025.116800.
- [4] ASHRAE, ANSI/ASHRAE/IES Standard 90.1-2022: Energy standard for buildings except low-rise residential buildings, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA (2022).

- [5] J. Lyu, D. Lai, H. Zhang, Z. Lian, LLM-powered agent for occupant-centric indoor environment control: Design, implementation, and evaluation, *Energy and Built Environment*In press (2026). doi:10.1016/j.enbenv.2026.04.001.
- [6] T. He, F. Jazizadeh, Context-aware LLM-based AI agents for human-centered energy management systems in smart buildings, arXiv:2512.25055 (2025).
URL <https://arxiv.org/abs/2512.25055>
- [7] Z. Qi, Z. Yang, M. Liu, Z. O'Neill, Energy-efficient HVAC control in residential buildings through occupant intention inference with multimodal large language models, *Advances in Applied Energy* 23 (2026) 100283. doi:10.1016/j.adapen.2026.100283.
- [8] W. Jung, Multi-agent home energy management assistant (HEMA), *SoftwareX* 34 (2026) 102633. doi:10.1016/j.softx.2026.102633.
URL <https://arxiv.org/abs/2602.15219>
- [9] W. Jung, K. Jeon, P. Babon-Ayeng, What you know does not show: Domain knowledge and occupant-AI interaction in building energy management, in: *Proceedings of the 2026 ACM Sustainability Week, 2026*, pp. 501–508. doi:10.1145/3765611.3815499.
- [10] L. Nicholas, S. Gauthier, E. Gerding, S. Stein, Towards generative occupants: LLM agents in physics-based building simulation, in: *ACM Sustainability Week Companion '26*, Banff, AB, Canada, 2026. doi:10.1145/3765611.3813753.
- [11] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, M. S. Bernstein, Generative agents: Interactive simulacra of human behavior, in: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi:10.1145/3586183.3606763.
- [12] M. J. Lee, R. Zhang, LLM-driven generative agents for simulating occupant feedback in built environments, *Journal of Computing in Civil Engineering* 40 (4) (2026) 04026045. doi:10.1061/JCCEE5.CPENG-7365.
- [13] A. N. N. Rende, T. Yilmaz, Ö. Ulusoy, Negotiating comfort: Simulating personality-driven LLM agents in shared residential social networks, arXiv:2507.09657 (2025).
URL <https://arxiv.org/abs/2507.09657>

- [14] C. Chen, Ö. Karaduman, X. Kuang, Behavioral generative agents for energy operations, arXiv:2506.12664 (2025).
URL <https://arxiv.org/abs/2506.12664>
- [15] Z. Leng, et al., AgentSense: Virtual sensor data generation using LLM agents in simulated home environments, arXiv:2506.11773, AAAI 2026 (2025).
URL <https://arxiv.org/abs/2506.11773>
- [16] U.S. Bureau of Labor Statistics, American time use survey, 2022–2023 microdata, <https://www.bls.gov/tus/> (2024).
- [17] U.S. Department of Energy, EnergyPlus version 23.2.0, <https://energyplus.net> (2023).
- [18] U.S. Energy Information Administration, Residential energy consumption survey (RECS) 2020, <https://www.eia.gov/consumption/residential/> (2023).
- [19] S. Ramírez, FastAPI: Modern, fast web framework for building APIs with Python, <https://fastapi.tiangolo.com> (2023).
- [20] Anthropic, Model context protocol (MCP) specification, <https://modelcontextprotocol.io> (2024).
- [21] W. Jung, S. Witt, Understanding smart thermostat adoption: Housing, HVAC, and socio-economic traits in the U.S., Energy Reports 15 (2026). doi:10.1016/j.egyr.2026.109244.
- [22] S. Gyamfi, S. Krumdieck, Price, environment and security: Exploring multi-modal motivation in voluntary residential peak demand response, Energy Policy 39 (5) (2011) 2993–3004. doi:10.1016/j.enpol.2011.03.012.
- [23] S. Kim, S. Chang, W. Jung, L. Debs, S. E. Jung, Integrating social justice criteria into multiscale spatial modeling of energy burden, Energy Research & Social Science 134 (2026). doi:10.1016/j.erss.2026.104630.
- [24] W. Jung, Y. Mo, Adoption, use, and perception of room and personal comfort systems in U.S. households, Journal of Building Engineering 127 (2026). doi:10.1016/j.jobe.2026.116313.

- [25] M. H. Albadi, E. F. El-Saadany, A summary of demand response in electricity markets, *Electric Power Systems Research* 78 (11) (2008) 1989–1996. doi:10.1016/j.epsr.2008.04.002.
- [26] W. Jung, P. Babon-Ayeng, Large language model-driven context-aware eco-feedback generation and evaluation, *Energy and Buildings* (2026). doi:10.1016/j.enbuild.2026.118038.
- [27] K. Fisher, J. Gershuny, E. Altintas, A. H. Gauthier, Multinational time use study: User’s guide and documentation, Tech. rep., Centre for Time Use Research, Oxford, UK, <https://www.timeuse.org/mtus> (2013).
- [28] Eurostat, Harmonised European time use surveys — 2018 guidelines, Tech. rep., Publications Office of the European Union, Luxembourg, ISBN 978-92-76-01964-0; <https://doi.org/10.2785/926903> (2019).
- [29] P. O. Fanger, *Thermal Comfort: Analysis and Applications in Environmental Engineering*, Danish Technical Press, Copenhagen, 1970.
- [30] W. Jung, P. Babon-Ayeng, Language as a sensor: Narrative-enhanced thermal comfort modeling, in: *Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 2026. doi:10.1145/3744256.3812583.
- [31] P. Babon-Ayeng, W. Jung, Assessing indoor heat strain through occupant language: A framework for building thermal resilience, *Building and Environment* 304 (2026). doi:10.1016/j.buildenv.2026.115052.
- [32] U.S. Department of Energy, Prototype building models, <https://www.energycodes.gov/prototype-building-models> (2023).
- [33] W. Jung, What LLM-simulated users can and cannot tell us about conversational energy management systems, in: *Proceedings of the 2026 ACM Sustainability Week*, 2026, pp. 428–432. doi:10.1145/3765611.3815484.
- [34] R. El Makroum, S. Zwickl-Bernhard, L. Kranzl, Agentic AI home energy management system: A large language model framework for residential load scheduling, *Results in Engineering* 29 (2026) 109857. doi:10.1016/j.rineng.2026.109857.
- [35] U.S. Department of Energy, The ecobee smart thermostat dataset (2017), <https://www.osti.gov/dataexplorer/biblio/dataset/1854924> (2021).

Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research

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Abstract

Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant’s personal history, current context, and the type of energy signal being delivered. This study presents BUILD OCC, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset covering 16,684 respondents. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface—Python library, REST API, and Model Context Protocol server—so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers ~~confirm~~ show that ATUS-grounded sampling ~~produces~~ reproduces empirically calibrated activity distributions and that demographic priors propagate into ~~coherent~~, persona-consistent agent reasoning across timesteps, establishing internal consistency across strata. BUILD OCC ~~is expected to advance how the human behavioral layer is modeled in building energy research, giving the community a shared foundation for more realistic and demographically representative studies of occupant-driven energy use~~ provides the building energy community with a reusable, openly available implementation of the occupant behavioral layer. BUILD OCC is openly released at <https://doi.org/10.5281/zenodo>.

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21192895 under the Apache License 2.0 and installable via `pip install buildocc`.

Keywords: Artificial Intelligence Agent, Occupant Behavior Modeling, Building Energy Use and Management, American Time Use Survey

Required Metadata

Current code version

Table 1: Code metadata (mandatory)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0
C2	Permanent link to code/repository	https://github.com/humanbuildingsynergy/BuildOcc
C3	Permanent link to reproducible capsule	https://doi.org/10.5281/zenodo.21192895
C4	Legal code license	Apache License 2.0
C5	Code versioning system	Git
C6	Software code languages, tools, and services used	Python 3.11+
C7	Compilation requirements, operating environments & dependencies	<code>pip install buildocc</code> ; requires an LLM API key (Anthropic, OpenAI, Google), or local Ollama
C8	Developer documentation	https://github.com/humanbuildingsynergy/BuildOcc/blob/main/README.md
C9	Support email for questions	wooyoungjung@arizona.edu

1. Motivation and Significance

Humans are complex systems that respond to environmental, social, and economic signals in ways that are difficult to predict, introducing substantial uncertainty in building energy use and management [1, 2]. This long-standing challenge has motivated the development of occupant behavior models that can capture the stochastic and adaptive nature of human decision-making [3].

However, most existing models rely on fixed schedules (e.g., ASHRAE 90.1 reference schedules [4]) or stochastic processes that do not comprehensively capture and reason about the occupant’s personal history, current context, or the type of energy signal being delivered [3, 2]. Consequently, these models cannot replicate the nuanced and adaptive responses that real occupants exhibit in response to various conditions.

The emergence of large language models (LLMs) and their architecture for reasoning and memory has opened new opportunities for artificial intelligence (AI) agents to simulate human-like decision-making processes. These agents are capable of contextual reasoning, memory retention, and adaptive behavior, making them suitable for modeling complex systems, including occupants in buildings. Recent studies demonstrated the potential of LLM-integrated agents to simulate occupant behavior in building energy contexts, showing that they can produce more realistic and adaptive responses to various energy signals compared to traditional models [3, 5, 6, 7, 8][3, 5, 6, 7, 8, 9]. Specifically, Nicholas et al. [10] proposed agents that adopt the generative agent memory architecture of Park et al. [11]—combining a persona-based core memory, a timestep-level memory stream, and periodic reflection—and demonstrate that this representation produces qualitatively different energy demand patterns than fixed-schedule or rule-based occupant models. Other recent approaches include agents conditioned on IEQ sensor readings for comfort and health feedback [12], personality-driven agents for negotiating shared thermostat setpoints [13], and a Thought-Action-Reflection-Journal framework for modeling energy behavior under dynamic pricing [14]. However, these agents are instantiated from hand-crafted profiles or small convenience samples [15]. In addition, ~~none provides~~ few provide quantitative validation against empirical activity benchmarks, and ~~none releases~~ few release an open-source platform for integration with building energy tools—leaving the community without a ~~general-purposes~~ shared, survey-grounded ~~foundation~~ starting point for reproducible occupant behavior research.

This study aims to fill this gap by introducing BUILD OCC, an open-source platform for LLM-based occupant behavior simulation and building energy management. BUILD OCC grounds agent personas and activity schedules in the American Time Use Survey (ATUS) [16], a nationally representative dataset covering tens of thousands of diary respondents across income, employment, and household-type strata. Each agent maintains a memory stream that accumulates timestep-level observations and periodically synthesizes higher-order behavioral insights through reflection—enabling coherent, adaptive decision-making across a simulation day rather than independent per-timestep responses. Users can drive agents through building simulation timesteps using any supported LLM provider and receive struc-

50 tured behavioral decisions—thermostat adjustments, device control actions, and demand response reactions—that feed directly into EnergyPlus [17] co-simulation, Home Assistant automations, or custom building management systems through a three-layer interface (Python library, REST API, and MCP server). Abstract base classes and a plugin registry allow the community to extend the platform with new occupant strata, behavioral data
55 sources, and memory architectures. Consequently, BUILD OCC ~~is positioned to advance how the human behavioral layer is modeled in building energy research, giving the community a shared foundation for more realistic and demographically representative studies of occupant-driven energy use~~
60 a reusable and extensible implementation of the occupant behavioral layer, released openly so that it can be evaluated and built upon by other research groups.

2. Software Description

Following an overview of the platform architecture, the platform is described
65 in ~~five~~ six parts: the ATUS behavioral grounding that drives activity scheduling and persona sampling; the inputs required at simulation initialization; the per-timestep LLM reasoning pipeline; the demand response signal interface; ~~and~~ the extensibility mechanisms that allow the platform to be adapted to new research contexts; and the computational cost of running it.

70 2.1. Architecture Overview

BUILD OCC is organized around two levels of architecture: an internal *agent component architecture* that defines how occupant agents perceive, reason, and act within a simulation timestep, and a *platform interface* that exposes this functionality to external building energy tools (Figure 1).

75 ~~Agent Component Architecture.~~

2.1.1. Agent Component Architecture

The BUILD OCC occupant agent is composed of six interacting components:

- **Persona** — a structured demographic profile assigned to one of four strata (O1–O4; Table 2), with appliance ownership, income bracket,
80 and thermal comfort tolerance sampled from RECS [18] probability distributions.
- **ActivityScheduler** — samples the occupant’s activity at each 15-minute timestep from stratum- and hour-specific probability distributions derived from ATUS 2022–2023 diary records (16,684 respondents),
85 with weekday/weekend stratification (Appendix A).

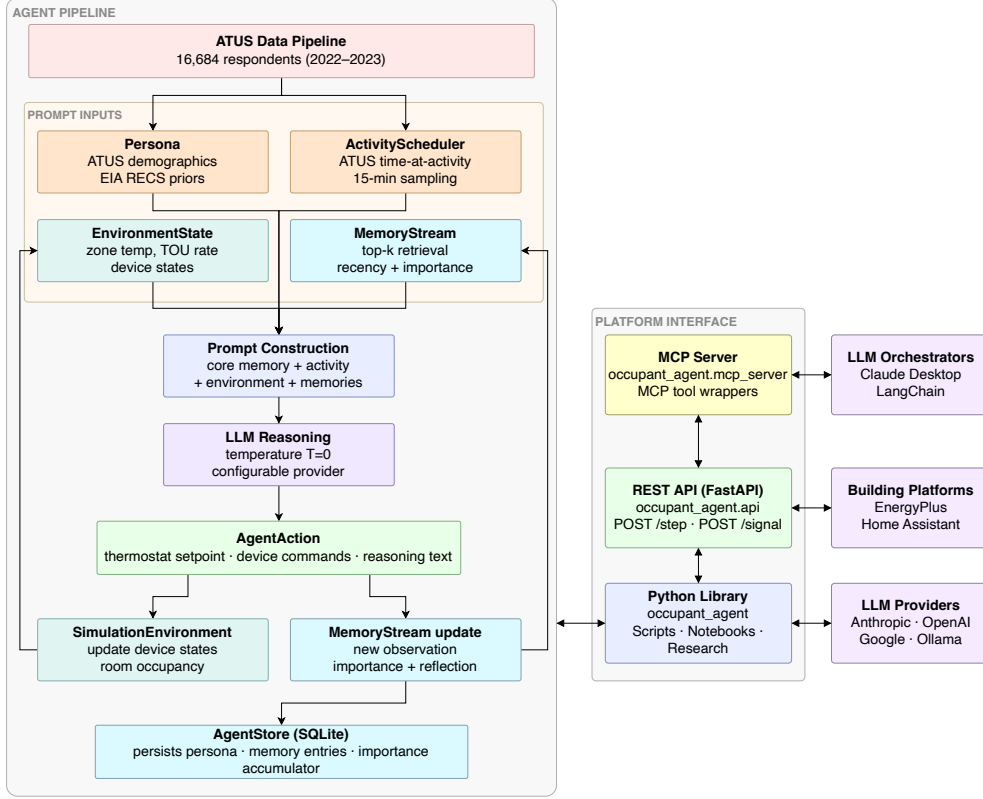


Figure 1: Architecture of BUILD OCC. *Left*: per-timestep agent pipeline showing how ATUS-grounded activity codes, retrieved memories, and environment state are assembled into an LLM prompt and how the resulting **AgentAction** updates both the simulation environment and the memory stream. *Right*: three-layer platform interface through which external building energy tools and LLM orchestration frameworks access the agent.

- **MemoryStream** — an accumulating log of observations, actions, and reflections ranked by recency and LLM-assigned importance following Park et al. [11]; top- k entries are injected into each timestep prompt and periodically synthesized into higher-order behavioral insights through a reflection call (Appendix B).
- **LLM Reasoning Engine** — assembles a prompt from the persona description, current ATUS activity, environment state, and retrieved memories, then queries a configurable LLM provider (Anthropic, OpenAI, Google, or local Ollama) at temperature $T=0$ for reproducibility, returning a structured action with a natural-language reasoning string.

- **SimulationEnvironment** — maintains ~~the state of the physical building between timesteps: device on/off device~~ states, room occupancy, and thermostat setpoint ~~. At between timesteps: at~~ the start of each timestep ~~;~~ it exposes the current state to the LLM Reasoning Engine ~~as the environment observation;~~ ~~and~~ at the end ~~;~~ it applies the agent’s returned action ~~and updates its internal state for the next timestep.~~
- **AgentStore** — persists the agent’s cognitive state ~~— persona, all (persona,~~ memory entries, and the importance accumulator ~~—)~~ to SQLite via SQLAlchemy after each operation. Environment state is owned by the caller and passed fresh with each request, enabling a stateless REST API pattern that supports multi-process deployment ~~without additional state management.~~

2.1.2. Platform Interface

BUILD OCC is distributed as the package `builddocc` and imported as `occupant_agent`.
 110 It exposes the agent through three integration layers, each targeting a different class of integrator:

1. **Python library** (`occupant_agent`) — ~~provides in-process Python~~ access to all agent components with no server overhead, ~~making it the recommended choice;~~ the recommended path for batch simulations ~~;~~ ~~Jupyter notebook experiments, and research workflows that need direct API access or custom component construction and notebook experiments.~~
2. **REST API** (`occupant_agent.api`) — a stateless FastAPI [19] server exposing ~~four HTTP operations:~~ agent initialization, per-timestep **behavioral** decisions (POST `/step`), ~~demand response~~ signal delivery (POST `/signal`), and state query (GET `/state`). ~~Suitable;~~ suitable for EnergyPlus co-simulation callbacks and any **building** platform that can issue HTTP requests.
3. **MCP server** (`occupant_agent.mcp_server`) — wraps the REST API as ~~tools conforming to the~~ Model Context Protocol (MCP) [20], ~~a standard interface that allows LLM applications to invoke external services. Any [20] tools, so any~~ MCP-compatible host ~~— (Claude Desktop, LangChain, or Home Assistant —~~ Home Assistant) can drive a ~~BUILD OCC simulation without writing~~ simulation without additional integration code.

2.2. ATUS Behavioral Grounding

Activity schedules are derived from ATUS 2022–2023 microdata (16,684 respondents). Time-at-activity probabilities are aggregated across eight categories (sleeping, work, food preparation, laundry, television viewing, eating,

exercise, and other) at 15-minute resolution, stratified by demographic group and day type (weekday / weekend). The method follows the “what is the person doing at this moment?” framing: for each diary episode $[t_{\text{start}}, t_{\text{stop}}]$, the method queries whether the episode covers the half-hour mark of each hour h , then computes a weighted probability across respondents using ATUS person-level weights. ATUS records a single primary activity per episode, so simultaneous secondary activities—watching television while preparing food, for example—are not represented in the scheduler’s output.

Four demographic strata are defined (Table 2), selected to span the variation in building occupancy and demand response participation that is most relevant to residential energy research (Appendix C).

Table 2: Demographic strata

ID	Label	ATUS filter
O1	Employed single adult	Full-time employed, lives alone, age 25–44
O2	Retired couple	Age 65+, not employed
O3	Employed parent	Full-time employed, children present, age 35–54
O4	Unemployed adult	Not employed, age 25–44

An important methodological choice is the *in-progress* framing: rather than sampling from the start-time distribution (which biases toward short activities), the method evaluates which activity is in progress at the half-hour mark of each hour, reflecting the actual time spent in each activity state rather than the frequency of starts. The formal probability computation is described in Appendix A.

Work-from-home (WFH) behavior is handled separately: a per-agent WFH probability encodes the fraction of working days spent at home (drawn from ATUS 2022–2023 telework questions), and a per-day Bernoulli draw determines WFH status for each simulated day. The WFH flag is included in the agent’s per-timestep prompt so that the LLM reflects both the ATUS activity code and the work location, preventing the scheduling model from assigning “work” activities that place the agent outside the home on days it is working remotely.

Each stratum uses stratum-specific appliance ownership priors from EIA RECS ~~[18]~~ [18, 21] and an income-bracket-modulated comfort tolerance (in °C), reflecting the documented tendency of lower-income households to accept wider thermal deviations to reduce HVAC costs ~~[22, 18]~~ [22, 18, 23, 24]. Income brackets are currently sampled uniformly within each stratum’s RECS income range.

2.3. User-Provided Inputs

Table 3 lists all inputs required at initialization; the only per-timestep value the calling application must supply is zone temperature, since it depends on the building’s physical response to occupant actions. To lower the barrier to first use, the library ships preset helpers for a representative household device inventory and room layout. A command-line interface also ships preset helpers and a YAML-driven configuration path for users who prefer not to write Python.

Table 3: User-provided inputs to BUILD OCC. All are specified once at initialization, except zone temperature, which the calling platform supplies at each timestep.

Input	Type	Description
Stratum	String	Demographic profile: O1–O4
Seed	Integer	Controls persona attribute sampling, activity scheduling, and work-from-home day draws; the same seed produces an identical agent across runs.
LLM provider	String	Anthropic, OpenAI, Google, or Ollama
Device inventory	List	Appliance IDs, initial on/off states, rated power (W)
Room list	List	Rooms the occupant can occupy
Thermostat setpoint	Float	Initial comfort setpoint (°C)
Zone temperature	Source	Fixed value, CSV file, or live building platform
Outdoor temperature	Source	Fixed value, CSV file, EPW file, or callable
TOU rate schedule	Source	Fixed value, CSV file, or tariff function

Users without a live co-simulation link can supply a constant zone temperature ~~value (suitable for behavioral experiments)~~ (suitable where thermal dynamics are not the focus) or a pre-computed CSV schedule indexed by ~~datetime(e.g., from a prior building energy simulation run or thermostat export)~~; a sample file covering one week of EnergyPlus output is included in the repository. ~~Outdoor temperature is specified once at initialization; supported sources include a fixed value, a CSV schedule, a user-supplied callable, or an EnergyPlus Weather (EPW) file.~~

2.4. LLM Reasoning Pipeline

With all inputs initialized, at each 15-minute simulation timestep the agent executes the following five-step pipeline:

- 185 1. **Activity resolution:** the current ATUS code is translated to a natural-language description and an occupancy flag using a lookup table of ATUS activity codes.
2. **Memory retrieval:** up to five entries are selected by a retrieval score that combines recency (exponential decay, half-life 24 h) and a 1–10 importance score the LLM assigns to each entry at the time of recording
190 — ensuring that consequential events remain retrievable even as they age (see Appendix B), following Park et al. [11].
3. **Prompt construction:** a system prompt encodes the persona’s stable identity — demographics, appliance ownership, and comfort tolerance.
195 A user prompt encodes the current environment state, ATUS activity, and the retrieved memories.
4. **LLM call:** the prompt is sent to a configurable provider (Anthropic Claude, OpenAI GPT, Google Gemini, or Ollama) at temperature $T=0$ for reproducibility. The model selects from four action types (Table 4),
200 returning a JSON object with target, value, and a one-sentence memory note.
5. **Memory update:** the observation and its LLM-assigned importance score are appended to the stream. If the cumulative accumulator exceeds a threshold (default 100), a reflection pass is triggered in which
205 a higher-capability model distills recent observations into three higher-order insights stored as reflection entries.

Table 4: The four action types returned by `step()` at each timestep, with their decision trigger and a representative example. Demand response is handled separately via `receive_signal()` (Section 2.5).

Action type	Trigger	Example
<code>do_nothing</code>	No environmental or activity condition warrants intervention	Agent is sleeping; no device is running
<code>adjust_thermostat</code>	Zone temperature deviates from comfort tolerance, or TOU rate context	Raises setpoint during peak window to reduce HVAC cost
<code>toggle_device</code>	Activity–appliance alignment (on) or inactivity (off)	Turns off lights when leaving for work
<code>move_room</code>	Activity context suggests transitioning to a different room	Moves to bedroom at 23:00 for sleeping

Setting the sampling temperature to $T=0$ makes the model return its single most probable output rather than sampling among alternatives, so a fixed provider, model version, and prompt reproduce the same action on every run. This determinism does not extend across providers, or across successive versions of a hosted model, because it is relative to one specific set of model weights. Reproducible studies therefore record the provider and model version alongside the seed, and a locally hosted open-weight model can be pinned where results must remain reproducible over time.

The pipeline above governs autonomous per-timestep behavior; demand response scenarios additionally require the ability to deliver external signals to the agent, which is handled by the interface described in the following subsection.

2.5. Demand Response Signal Interface

Alongside the per-timestep pipeline, BUILD OCC provides a dedicated signal delivery interface that can be invoked at any point in a simulation. Three signal types are supported (Table 5), following established behavioral intervention typologies [22, 25][22, 25, 26]. Signals are delivered by passing a signal type, a free-text message, and the current environment state to the agent, which returns a response code (*accepted*, *rejected*, or *deferred*) alongside a natural-language reasoning string drawn from the agent’s memory context. Keeping the three types mutually exclusive enables factorial experimental designs in which signal type and demographic stratum are the primary independent variables.

Table 5: Demand response signal types supported by BUILD OCC, classified by behavioral lever and illustrated with representative example messages. All three types share the same delivery mechanism and response codes (*accepted*, *rejected*, *deferred*).

Type	Behavioral lever	Example message
A — Direct command	Regulatory compliance	“Turn off your HVAC for 30 min to support grid stability during today’s peak demand event.”
B — Educational	Economic information	“Your HVAC costs $3\times$ more per kWh at the current peak rate — raising the setpoint 1°C saves an estimated \$0.35 today.”
C — Social norm	Peer comparison	“75% of similar households in your area have reduced HVAC use during today’s peak event.”

230 The extensibility mechanisms that allow all of these components to be adapted
for new research contexts are described in the following subsection.

2.6. Platform Extensibility

BUILD OCC is designed for adaptation across research contexts without mod-
ifying the core library. Extensibility is organized along three independent
235 axes:

- **Occupant population** (`BasePersona`): new demographic profiles — commercial occupants, hotel guests, non-US populations, or any custom stratum — are registered by implementing `BasePersona`.
- **Activity grounding** (`BaseScheduler`): the built-in scheduler uses
240 ATUS microdata; any other grounding source (a Markov chain, a physics-based occupancy model, or alternative activity data) can be substituted by implementing `BaseScheduler`.
- **Memory architecture** (`BaseMemoryStream`): the built-in recency-plus-importance retrieval can be replaced with embedding-based retrieval or any other scheme by subclassing `BaseMemoryStream`.
245

Extensions are registered via Python’s package entry-point mechanism and discovered automatically at import time, allowing third-party packages to extend the platform without forking the repository:

Listing 1: pyproject.toml entry-point declarations for a third-party stratum and scheduler.

```
250 1 [project.entry-points."occupant_agent.strata"]  
2   P5 = "my_package.personas:LowIncomeElderlyAlone"  
3  
4   [project.entry-points."occupant_agent.schedulers"]  
255 5   homer = "my_package.schedulers:HomerScheduler"
```

260 Transfer to another country is a data-substitution task rather than a redesign: the scheduler consumes a table of activity probabilities by stratum, hour, and day type, so any national time-use survey following harmonized diary conventions—the Multinational Time Use Study [27] or the European HETUS series [28], for example—can be substituted by regenerating that table behind `BaseScheduler`. What does not transfer automatically is the persona layer, whose appliance-ownership and income priors come from a US survey (RECS) and would need a national equivalent.

Researchers can inject additional context into the LLM reasoning pipeline
265 — such as predicted mean vote (PMV) comfort indices [29], language-based comfort and heat-strain signals [30, 31], or indoor air quality readings —

either by extending the agent class or by passing a free-text string at each timestep; both paths are available through the REST API and MCP server. The environment state and signal schemas are fixed to ensure reproducibility, and a test utilities module provides mock implementations and conformance tests for extension authors.

2.7. *Computational Cost and Scalability*

Because the agent queries a language model at nearly every timestep, cost and runtime scale linearly with simulated agent-days. A 15-minute timestep gives up to 96 calls per simulated day per agent, plus occasional reflection calls, with responses capped at 512 tokens; the monetary cost therefore follows directly from the provider’s per-token price at the time of use, and wall-clock time is dominated by provider latency rather than local computation. Two mechanisms reduce this. Any local open-weight model served through Ollama removes per-call cost entirely, at the price of reduced reasoning quality, and scheduler-only studies—including the Tier 1 validation reported in Section 3.2—require no model calls at all. Cost, rather than platform capability, is what bounded the simulation lengths reported in Section 3.

These figures are per agent. Because agents are independent—each carries its own persona, memory, and schedule, and the REST API is stateless—a multi-occupant household is simulated by running one agent per occupant, so call volume scales linearly with household size (a four-occupant household is roughly 384 calls per simulated day) while the agents themselves can be run in parallel across processes. Interaction between occupants, such as negotiation over a shared thermostat, is not modeled in this version and is identified as future work in Section 5.

3. Illustrative Examples

This section demonstrates the platform in three parts: a walkthrough of a single agent through a demand response event, followed by two quantitative tiers. Tier 1 validates the *ActivityScheduler*, testing whether ATUS-grounded sampling produces category distributions matching empirical ATUS reference data relative to a deterministic rule-based baseline; Tier 2 validates the *LLM agent decisions*, testing whether each stratum’s demographic priors propagate into distinguishable behavioral patterns under identical environment conditions. Both are necessary because a well-grounded scheduler and a well-calibrated reasoning agent are independent failure modes — one can be correct while the other is not.

3.1. Demand response walkthrough

To illustrate agent behavior during a residential demand response event, this ~~section~~ subsection traces an O1 agent (28-year-old employed male, income bracket 9, comfort band $\pm 1.1^\circ\text{C}$) through the peak electricity period of a summer weekday evening (18:00–19:45, TOU rate \$0.22/kWh). Zone temperatures come from pre-computed EnergyPlus output for the DOE PNNL residential prototype building [32] (single-family, Climate Zone 2B, variable-speed heat pump) included in the repository. The simulation loop is:

Listing 2: Running the BuildOcc simulation loop with the sample zone temperature CSV.

```
1 # Full imports available in examples/simulation_loop.py (
    repository)
2 from datetime import datetime, timedelta
3
315 4 agent      = OccupantAgent.from_stratum("O1", seed=42)
5 scheduler = ActivityScheduler(stratum="O1", seed=42)
6 zone_temp = zone_temp_from_csv(...) # pre-computed
    EnergyPlus zone temperatures
7
320 8 sim = SimulationEnvironment(
9     initial_devices=persona_devices(agent.persona),
10    initial_rooms=[RoomState(room_id="living_room",
        occupied=True)],
325 11 thermostat_setpoint=22.0,
12    outdoor_temp_fn=summer_day_temp,
13    tou_rate_fn=peak_tou_rate,
14 )
15
330 16 base = datetime(2025, 8, 11, 18, 0)
17 for i in range(8):
18     ts      = base + timedelta(minutes=15 * i)
19     env      = sim.observe(ts, zone_temp(ts))
20     action   = agent.step(env, atus_code=scheduler.sample(
335    ts))
21     sim.apply(action, ts)
22     print(ts.strftime("%H:%M"), action.action_type,
        action.reasoning)
```

340 At 18:00 (Step 1), the CSV-sourced zone temperature is 23.9°C and the thermostat is set to 22°C . With outdoor temperatures exceeding 38°C , the HVAC is running continuously to hold the setpoint. The agent is engaged in computer leisure and proactively raises the thermostat setpoint to 24°C to reduce peak-rate HVAC run time, reasoning:

345 *“The indoor temp is 23.9° C and the thermostat is set to 22° C —
the HVAC is running hard to maintain the setpoint with 39° C
outside during peak rate hours (\$0.22/kWh). I’m comfortable at
23.9° C, well within my 1.1° C tolerance, so raising the setpoint
to 24° C will meaningfully cut cooling costs during this expensive
350 period.”*

This response demonstrates that the persona’s income bracket and comfort tolerance surface in actionable decision-making without explicit rule encoding.

At 18:30 (Step 3), a Type B educational signal is delivered: *“Your HVAC is
355 running during peak hours (4–9pm) at \$0.22/kWh—three times the off-peak
rate. Raising the setpoint by 1° C saves ~\$0.35 today.”* The agent accepts the signal but references its own prior action stored in memory:

360 *“I already raised the thermostat to 24° C earlier today and it’s
working—I’m at 23.9° C and comfortable within my tolerance.
The math is clear (\$0.35 savings) and the effort is zero since
I’ve already taken this action.”*

This cross-timestep ~~memory coherence—where the agent recalls and references~~
~~coherence—the agent recalling and acting on~~ its own prior ~~decision—illustrates~~
~~the emergent behavioral consistency that distinguishes LLM-grounded agents~~
365 ~~from rule-based or lookup-table baselines. A fixed schedule model could~~
~~encode a thermostat raise at 18:00, but it cannot encode the nuance that the~~
~~agent chose that action for cost reasons and remembers having already acted~~
~~when a subsequent signal arrives~~~~decision—is what a fixed schedule cannot~~
~~represent.~~

370 Over the remainder of the evening, ~~the agent maintains the raised setpoint~~
~~without further intervention. Importance; importance~~ ratings for routine
observations ~~are stay~~ low (2–3 out of 10); ~~correctly prioritizing memory~~
~~budget toward the salient cost-saving event. By contrast, an O4 agent~~
~~(unemployed adult) carries a fuller day-long memory of peak-rate exposure~~
375 ~~and, in representative runs, accepts the same Type B signal citing consistency~~
~~with prior device-toggling decisions rather than comfort margin—a qualitatively~~
~~different rationale that reflects demographic grounding rather than explicit~~
~~rule differences.~~

Figure 3 illustrates the behavioral fidelity of the activity scheduler for stratum
380 O1 (employed single adult) on weekdays. The ATUS-grounded simulation
(panel a) closely tracks the faint reference lines, while the fixed rule-based
baseline (panel b) diverges markedly—most notably collapsing all daytime

activity into sharp step functions and failing to reproduce the gradual evening wind-down.

Hourly activity probability distributions for stratum O1 (employed single adult, weekday) across five activity categories. (a) ATUS-grounded simulation (180 simulated days): bold lines closely track the reference, confirming scheduler fidelity. (b) Fixed rule-based baseline: bold lines diverge sharply, with the baseline collapsing smooth activity transitions into unrealistic step functions and misassigning the O1 late-evening activity peak.

Validation

Validation is organized in two tiers targeting different components of the platform. Tier 1 validates the *ActivityScheduler*: it tests whether ATUS-grounded activity sampling produces category distributions that match empirical ATUS reference data, relative to a deterministic rule-based baseline. Tier 2 validates the *LLM agent decisions*: it tests whether the demographic priors encoded in each stratum’s persona propagate into distinguishable behavioral patterns under identical environment conditions. Both tiers are necessary because a well-grounded scheduler and a well-calibrated reasoning agent are independent failure modes—one can be correct while the other is not.

3.2. Tier 1: Activity Schedule Fidelity

The ATUS-grounded activity scheduler is validated against a deterministic rule-based baseline using KL divergence between simulated activity distributions and ATUS empirical distributions. For each stratum, 180 days of 96 timesteps each (17,280 observations per stratum) are simulated, the empirical category distribution at each hour is computed, and compared to the ATUS reference using:

$$D_{\text{KL}}(P\|Q) = \sum_c P(c) \log \frac{P(c)}{Q(c) + \varepsilon} \quad (1)$$

where P is the ATUS reference distribution, Q is the simulated distribution, and $\varepsilon = 10^{-9}$ is a smoothing constant. Table 6 and Figure 2 report the mean per-hour KL divergence averaged across all 24 hours and both day types. Figure 3 (preceding section) shows the corresponding hourly distributions for O1, illustrating how closely the ATUS-grounded scheduler tracks the empirical reference compared to the fixed baseline.

The fixed baseline assigns deterministic activities by hour-of-day rules (e.g., sleeping 23:00–06:59, work 09:00–16:59) without demographic conditioning, which produces large divergence against the ATUS reference—especially for the retired (O2) and unemployed (O4) strata, whose schedules differ most from the employed-adult assumption. The 1,224-fold reduction in KL divergence

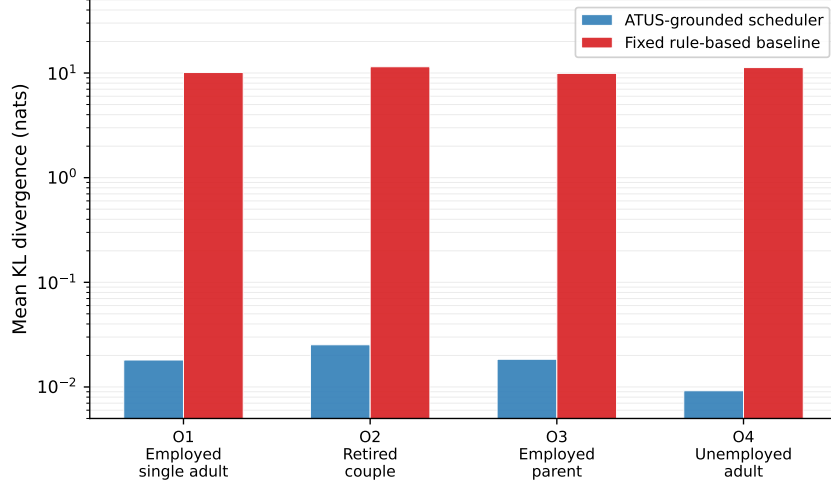


Figure 2: Mean per-hour KL divergence (nats, log scale) between simulated and ATUS empirical activity distributions over 180 simulated days. ~~Annotated ratios indicate the fold-reduction achieved. The baseline’s magnitude is governed by the ATUS-grounded scheduler relative to smoothing constant, since it places zero probability on categories the fixed-rule-based baseline reference records as occurring.~~

Table 6: Mean per-hour KL divergence (nats): ATUS-grounded vs. rule-based scheduler. Computed over 180 simulated days, 96 timesteps each, weekday/weekend split. Lower is better.

Stratum	ATUS scheduler	Fixed baseline
O1 (Employed single adult)	0.0181	10.11
O2 (Retired couple)	0.0253	11.48
O3 (Employed parent)	0.0184	9.92
O4 (Unemployed adult)	0.0092	11.27

for O4 reflects the extreme mismatch between a generic nine-to-five work rule and the ATUS-observed daytime activity patterns of unemployed adults. Every ATUS-grounded condition stays below 0.03 nats, while the fixed baseline ranges from 9.92 to 11.48. The gap is one of kind rather than degree: the baseline places zero probability on categories the reference records as occurring, so its divergence is governed by the smoothing constant ϵ and registers the presence of impossible predictions rather than the size of a behavioral error.

Figure 3 illustrates what this divergence looks like category by category for stratum O1 on weekdays: the ATUS-grounded simulation (panel a) closely tracks the reference distributions, while the fixed rule-based baseline (panel

430 b) diverges markedly—most notably collapsing daytime activity into sharp step functions and failing to reproduce the gradual evening wind-down.

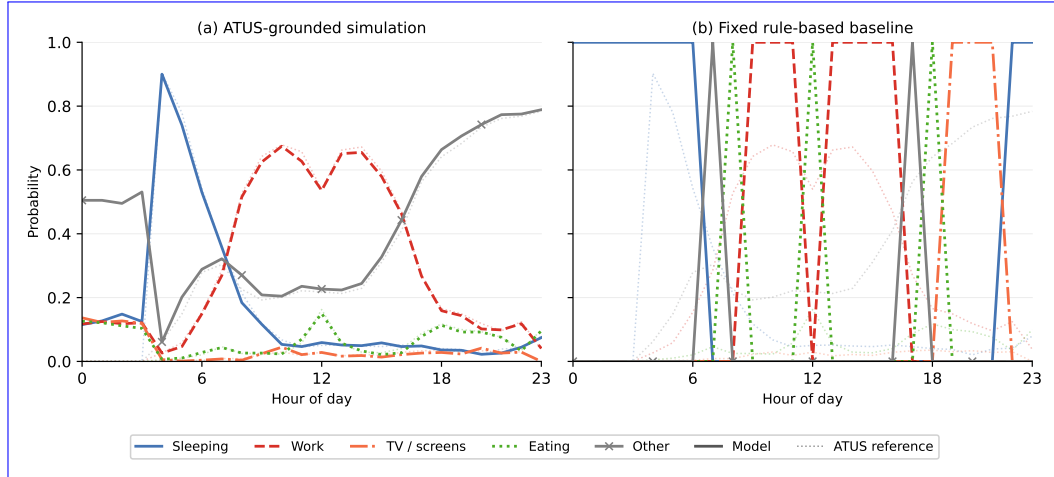


Figure 3: Hourly activity probability distributions for stratum O1 (employed single adult, weekday) across five activity categories. (a) ATUS-grounded simulation (180 simulated days): bold lines closely track the reference. (b) Fixed rule-based baseline: bold lines diverge sharply, with the baseline collapsing smooth activity transitions into unrealistic step functions and misassigning the O1 late-evening activity peak.

3.3. Tier 2: Persona-Driven Behavioral Differentiation

Tier 2 tests whether the demographic priors encoded in each stratum’s persona produce distinguishable behavioral patterns under identical environment conditions, ~~confirming that persona variation—not environment variation—is the primary driver of the observed differences~~so that any observed differences are attributable to persona variation rather than environment variation. Each of O1–O4 was run for three independent seeds over one weekday (96 timesteps at 15-minute resolution) under identical environment conditions—the same EnergyPlus zone temperature, device configuration (HVAC, three lighting zones, TV, washer, dishwasher), and time-of-use (TOU) tariff (\$0.22/kWh peak, \$0.08/kWh off-peak)—~~so that behavioral differences are attributable solely to persona variation.~~

Table 7 reports the resulting action-type distributions and mean thermostat setpoints during the TOU peak window (4–9 pm). O2 (retired couple) ~~exhibits the highest overall activity level—lowest inactivity rate~~is the most active stratum—lowest inactivity (67.4%), highest device-toggling rate (8.0%) ~~highest and~~ room-movement rate (21.5%) ~~and rates, and the~~ highest mean peak setpoint (24.3°C)—consistent with full-day home occupancy ~~and~~

active use of three configurable lighting zones as the agent transitions between kitchen, dining room, and living room throughout the day. O4 (unemployed adult; ATUS $n=107$, Appendix C) matches O2’s toggling rate (8.0%) but records the lowest thermostat-adjustment rate (1.4%) and the lowest peak setpoint (23.1°C), suggesting cost-sensitive passive thermostat management despite active device engagement. Because O4’s probability tables derive from only 107 ATUS respondents—the smallest of the four strata and the noisiest at 15-minute resolution—this pattern should be read as provisional; it may reflect sampling noise in the grounding data as much as a demographic tendency. O3 (employed parent) shows the lowest device-toggling rate (3.5%); its peak-hour schedule dominated by ATUS child-care codes (physical care, play, and supervision activities) dominate O3’s peak-hour schedule, reducing behavioral attention available for appliance and thermostat control, while O1 (employed single adult) sustains a moderate profile across all action types, with an inactivity rate of 76.0% and the second-highest peak setpoint (23.5°C) throughout. Figure 4 visualizes these distributions.

Table 7: Action-type distribution (%) per stratum: mean over three seeds, one weekday each (96 steps), giving 288 timesteps per stratum. At this sample size one percentage point corresponds to roughly three timesteps, so differences of a few tenths should not be read as meaningful. Peak SP is the mean thermostat setpoint during TOU peak hours (4–9 pm, \$0.22/kWh).

Stratum	Inactive	Thermostat	Toggle	Move	Peak SP (°C)
O1 (Employed single)	76.0	2.8	4.2	17.0	23.5
O2 (Retired couple)	67.4	3.1	8.0	21.5	24.3
O3 (Employed parent)	76.4	3.1	3.5	17.0	23.6
O4 (Unemployed)	72.9	1.4	8.0	17.7	23.1

To assess demand response participation, three standardized signals were delivered to each stratum (five seeds each) after a 2.5-hour context period at peak-rate conditions (10 simulated timesteps beginning at 16:00, giving each agent an experiential baseline in memory before signal delivery): Type A (direct command: “Turn off your HVAC for 30 minutes”), Type B (educational price signal: cost comparison at three times the off-peak rate), and Type C (social norm: “75% of similar households have reduced HVAC use”). Table 8 reports accept rates the number of accepting agents per stratum and signal type. Direct commands (Type A) achieved the highest drew the most acceptance for O2 (retired couple, 40% 2 of 5 seeds), with O1 (employed single, 20%) and O4 (unemployed adult, 20%) showing moderate compliance) accepting in 1 of 5; educational price signals (Type B) matched this rate for were accepted by O2 and O4 (40% each), suggesting that in 2 of 5 seeds

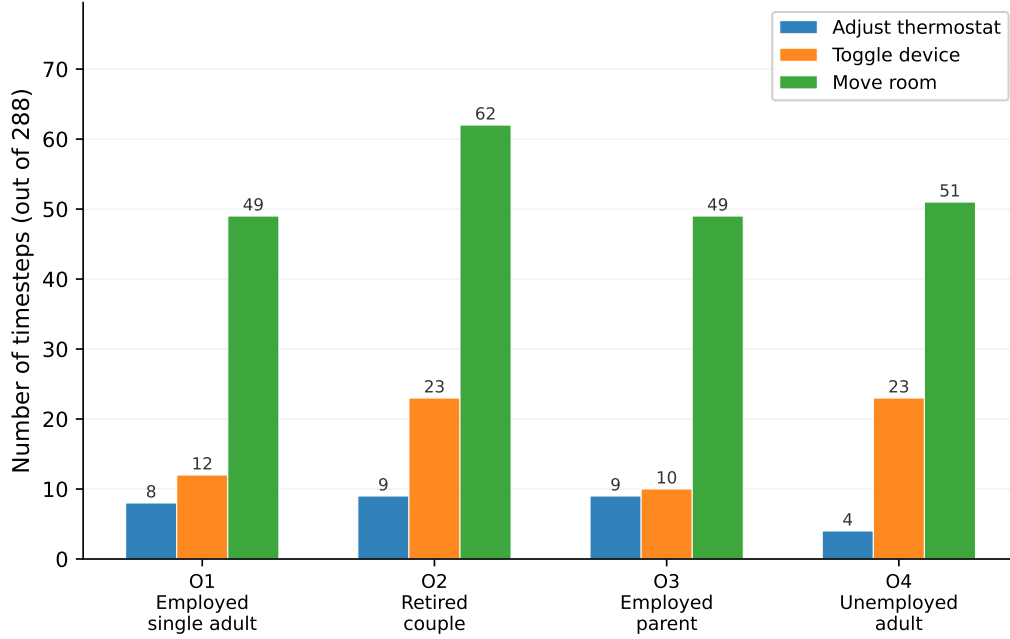


Figure 4: Active action counts per stratum across 288 timesteps (3 seeds $\times$ 96 steps, identical environment conditions). Do-nothing accounts for 67–76% of all timesteps across strata and is excluded to isolate cross-stratum differences in active decision-making. Bar labels show absolute counts.

each, a pattern consistent with cost-aware strata ~~respond~~ responding to both authority and economic framing. O3 (employed parent) rejected all signal types across all five seeds: signal delivery at 18:30 coincides with child-care activity codes in the ATUS schedule, making ~~the 0% acceptance rate that~~ outcome an expected consequence of schedule grounding rather than a general insensitivity to demand response. Social norm signals (Type C) ~~achieved~~ zero acceptance across all strata ~~draw no acceptance in any stratum~~, consistent with field evidence on the limited efficacy of peer-comparison messaging [22, 25]. With five runs per cell these counts indicate direction only; no significance testing is performed on them.

~~Figure ?? compares thermostat setpoint trajectories of O1 and O3 over the simulated day. O1 agents operate at setpoints averaging 0.5°C higher than O3 during peak hours; the single adult’s narrow comfort band (1.1°C) and absence of family obligations enable individual thermostat optimization; while O3’s schedule is dominated by child-care activities during the peak window.~~

Thermostat setpoint trajectories for O3 (employed parent, family-constrained,

Table 8: Signal compliance ~~rates (accept %)~~ per stratum and signal type ~~over~~, reported as ~~the number of accepting agents out of five random seeds. Percentages are not given, since five runs per cell cannot support them.~~ Type A: direct command; Type B: educational price signal; Type C: social norm nudge.

Stratum	Type A	Type B	Type C
O1 (Employed single)	20 <u>1/5</u>	0/5	0/5
O2 (Retired couple)	40 <u>2/5</u>	40 <u>2/5</u>	0/5
O3 (Employed parent)	0/5	0/5	0/5
O4 (Unemployed)	20 <u>1/5</u>	40 <u>2/5</u>	0/5

~~upper panel) and O1 (employed single adult, comfort-seeking, lower panel). Shaded region: TOU peak period (4–9 pm). O1 agents sustain setpoints averaging 0.5°C higher than O3 during peak hours, reflecting individual cost-avoidance unconstrained by family obligations.~~

~~These results confirm~~ These results show that the ATUS demographic priors propagate into qualitatively distinct LLM-driven behavioral patterns: employment status shapes thermostat adjustment frequency, household composition modulates signal compliance, and full-day home occupancy drives device-toggling behavior. ~~Two design~~ Three constraints bound the interpretation. First, ~~and most fundamentally, Tier 2 establishes~~ internal consistency: agents instantiated from different demographic priors behave differently, and in directions consistent with those priors. It does not establish behavioral realism, which would require comparison against measured occupant data and is left to future work [33]. Second, each stratum is run for only one weekday per seed (three seeds for action distributions, five for signals), which is sufficient to show directional patterns but not to establish statistical significance; multi-day designs are feasible but were constrained by API cost. ~~Second~~ Third, Tier 2 uses a single environment configuration (fixed weather, tariff structure, and household composition), which limits generalizability across different climate zones and occupancy contexts. The behavioral differences across strata also reflect a structural property of the platform: both the ATUS-grounded activity schedule and the LLM persona reasoning are conditioned on the same demographic stratum, a coupling that ensures internal consistency between the scheduling and decision-making components.

4. Impact

BUILD OCC ~~enables~~ lowers the barrier to four lines of research that ~~were previously impractical~~ are difficult to pursue with fixed-schedule occupant models:

Survey-grounded behavioral agents. BUILD OCC ~~makes packages~~ population-calibrated occupant agents ~~a shared community resource as a reusable component~~ rather than a per-project infrastructure investment. Research groups can ~~now introduce realistic,~~ introduce demographically differentiated occupant behavior into any building energy study without constructing a grounding pipeline from scratch. The ATUS-derived activity distributions and RECS-conditioned personas provide ~~an empirically validated behavioral a~~ documented, survey-grounded baseline that any researcher can adopt, extend, or cite as a reference condition—~~shifting what was previously a specialized capability into a reusable foundation available to the entire building energy community.~~

Demographic behavioral experiments. BUILD OCC’s combination of ATUS-grounded demographic strata, persona income brackets, and configurable intervention signals enables controlled factorial studies of occupant behavior. Unlike fixed-schedule models, each agent reasons from its demographic persona and accumulated memory at every timestep, producing adaptive responses that reflect personal history and current context rather than predetermined patterns. Because both the activity schedule and the agent’s reasoning are conditioned on the same demographic persona, researchers can hold environment and intervention constant while varying stratum — or vary the intervention while holding stratum constant — isolating the contribution of socioeconomic and life-stage factors to any behavioral outcome. The Tier 2 signal experiment (Table 8) illustrates this: Type B educational signals ~~achieved drew~~ equal or higher acceptance than direct commands for cost-sensitive strata (O2 and O4: ~~40% 2 of 5 seeds~~ each), while O1 (employed single adult) responded only to Type A direct commands (~~20% 1 of 5~~) and O3 (employed parent) rejected all signal types — ~~compliance gradients that fixed-schedule models cannot reproduce~~ differences in direction that a fixed schedule cannot produce at all, having no mechanism to respond to a signal, though with five seeds per cell they are indicative rather than precise. The signal types correspond to established intervention typologies in behavioral energy research [22, 25], making results directly comparable to field studies.

Platform-neutral integration. The MCP server layer means that any tool with MCP support—including Claude Desktop, LangChain, energy scheduling systems [34], and custom orchestration frameworks—can drive a building simulation with ~~realistic-demographically differentiated~~ occupant behavior without writing integration code. The REST API provides an equivalent path for EnergyPlus co-simulation and Home Assistant integrations [5, 8].

Community extensibility. The plugin registry and abstract base classes lower the barrier for researchers who want to add custom demographic strata,

565 alternative behavioral data sources or activity grounding pipelines, or novel
memory architectures. The conformance test suite enables rapid validation
of extensions before publication.

5. Conclusions

BUILD OCC delivers a survey-grounded, three-layer occupant agent platform
570 that the building energy community can use, extend, and build upon with-
out reconstructing the behavioral layer for each study. ~~Two validation tiers~~
~~confirm~~ Tier 1 confirms that ATUS grounding produces empirically cali-
brated activity distributions ~~and~~; Tier 2 shows that demographic priors prop-
agate into ~~meaningfully~~ differentiated agent reasoning ~~under environmental~~
575 ~~and intervention signals~~, though it does not by itself establish that the
resulting behavior is realistic.

~~The platform's deeper contribution is a reframing: occupant behavior shifts~~
~~from an uncontrolled source of uncertainty to~~ Beyond the implementation,
580 the platform treats occupant behavior as a structured, demographically pa-
rameterizable experimental variable rather than an uncontrolled source of
uncertainty. The ATUS-derived strata, frozen schema, and plugin registry
~~together are intended to~~ make that variable reproducible and extendable
across independent research groups ~~a property that has been absent from~~
~~the occupant behavior modeling literature.~~

585 Current limitations include a single-action constraint per timestep ~~and~~; ATUS-
grounded strata covering US demographic profiles only; activity grounding
restricted to primary activities, since ATUS records only one activity at a
time; and a memory-importance score that the agent assigns to itself and
then reuses for retrieval, without external calibration or a feedback path
590 that would correct a mis-scored entry. Subsequent phases extend the plat-
form with multi-agent household simulation, non-US occupancy data, and
external behavioral benchmarks including ecobee Smart Thermostat field
data [35] matched to the four ATUS strata.

CRedit Author Statement

595 **Wooyoung Jung:** Conceptualization, Methodology, Software, Validation,
Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares that there are no competing interests.

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A. ATUS Activity Probability Computation

605 This appendix provides the formal description of the activity probability computation summarized in Section 2.

Episode-overlap framing. ATUS diaries record each respondent’s day as a sequence of episodes, each with a start time t_s (minutes since midnight), stop time t_e , and a six-digit activity code in ATUS. For a 15-minute simulation slot centered at time τ , the scheduler identifies which activity episode was in progress at the midpoint $\tau + 7.5$ min:

$$c(r, \tau) = \text{activity code of respondent } r \text{ such that } t_s \leq \tau + 7.5 \text{ min} < t_e \quad (\text{A.1})$$

where r indexes individual diary respondents. This in-progress query avoids the start-time bias that would overrepresent short activities (e.g., a 5-minute meal counts only if the midpoint falls within it).

615 *Category probability.* Eight activity categories are defined (Table A.1). The probability of category k at hour h , for demographic stratum s and day type $d \in \{\text{weekday, weekend}\}$, is:

$$P(k \mid s, d, h) = \frac{\sum_{r \in R(s, d)} w_r \cdot \mathbf{1}[c(r, h:30) \in k]}{\sum_{r \in R(s, d)} w_r} \quad (\text{A.2})$$

where $R(s, d)$ is the set of diary respondents matching stratum s on day type d ; w_r is the ATUS person-level survey weight (provided by BLS to correct for sampling design); $\mathbf{1}[\cdot]$ is the indicator function that equals 1 when respondent r was engaged in a category- k activity at the half-hour mark $h:30$ of hour h ; and the denominator normalizes the weighted counts to probabilities. Evaluating Equation (A.2) for all $h \in \{0, \dots, 23\}$ and all 8 categories yields a 24×8 weighted probability table per stratum per day type, which is bundled with the installed package.

Table A.1: The eight broad activity categories used in the two-step ATUS sampling procedure, with their ATUS code filter, the number of Tier-3 codes mapped to each category, and representative subcategories.

Category	ATUS code filter	Codes	Representative subcategories
Sleeping	prefix 0101	3	Sleeping; Sleeplessness
Work	prefix 05	8	Work (main job); Work (other job); Socializing at job
Food preparation	prefix 0201	4	Food/drink preparation; Food presentation; Kitchen clean-up
Laundry	code 020202 exactly	1	Laundry
Television	code 120301 exactly	1	Watching TV
Eating	prefix 1101	2	Eating and drinking
Exercise	prefix 13	29	Aerobics; Basketball; Biking; Swimming; Weightlifting
Other	residual	172	Grooming; Reading; Computer use; Social communication; Travel

Code-level sampling. Once a category k is sampled from $P(k \mid s, d, h)$, a specific activity code is drawn from within that category proportional to the total diary minutes logged across all stratum respondents for each code. This gives common activities realistic relative frequencies (e.g., sleeping in a bed is drawn far more often than napping on a sofa, even though both map to the “sleeping” category).

Work-from-home probability. Finally, the WFH probability p_{WFH} for each stratum is estimated from the ATUS TEWHERE variable — the “where were you when you did this activity?” field — restricted to employed respondents who worked on their diary day. p_{WFH} equals the survey-weighted fraction of those respondents who reported working at home for any work episode, computed separately for each stratum. A per-day Bernoulli draw from p_{WFH} then determines whether the simulated agent works from home on a given day.

B. Memory Retrieval and Reflection Mechanics

This appendix provides the mathematical detail for the memory retrieval and reflection logic implemented in `MemoryStream`, following the generative agent architecture of Park et al. [11].

645 *Retrieval scoring.* At each simulation timestep t , every stored memory entry m receives a combined retrieval score $s(m, t) \in [0, 1]$ that balances how recent the entry is against how important it was judged to be when written:

$$s(m, t) = 0.5 \tilde{r}(m, t) + 0.5 \tilde{i}(m) \quad (\text{B.1})$$

where $\tilde{r}(m, t)$ is the normalized recency score and $\tilde{i}(m)$ is the normalized importance score, both in $[0, 1]$ and weighted equally.

The recency component decays exponentially with a 24-hour half-life:

$$\tilde{r}(m, t) = 2^{-(t-t_m)/24\text{h}} \quad (\text{B.2})$$

650 where t_m is the simulation time (in hours) at which entry m was written, and $(t - t_m)$ is the elapsed time in hours since writing. The score equals 1.0 at the moment of creation and halves every 24 simulation hours, so memories from the previous day retain roughly half their recency weight while week-old entries contribute near zero.

655 The importance component normalizes the integer score $i(m)$ assigned by the LLM at write time:

$$\tilde{i}(m) = \frac{i(m)}{10} \quad (\text{B.3})$$

where $i(m) \in \{1, \dots, 10\}$ is an integer on a scale from 1 (routine, nothing unusual) to 10 (rare, extreme situation), self-assigned by the agent’s LLM in the same JSON response that produces the behavioral action. The top- k entries ranked by $s(m, t)$ are injected into the next prompt ($k = 5$ by default).
660

Reflection trigger. An importance accumulator A tracks the cumulative importance of all entries written since the last reflection. After each entry m is stored, $A \leftarrow A + i(m)$. When $A \geq \theta$ (default $\theta = 100$), reflection is triggered: a dedicated LLM call receives the 30 most recent entries by simulation time
665 and returns exactly three synthesized behavioral insights (e.g., “this occupant pre-cools before peak hours to avoid peak tariffs”). Each insight is written back into the stream as a new entry with importance 9.0, and A resets to zero. At average importance 7–10 (high-salience events), reflection fires roughly every 10–14 timesteps (≈ 2.5 – 3.5 simulation hours); at average importance 2–3
670 (routine observations), every 33–50 timesteps (≈ 8 – 12 simulation hours).

C. Demographic Strata Selection Rationale

The four strata were selected to maximize variation in the two behavioral dimensions most consequential for residential demand response research: *peak-*

hour occupancy schedule and household composition. The selection is grounded
in the ATUS 2022–2023 microdata that drives the platform itself.
Table C.1 summarises the occupancy metrics and sample sizes derived from
ATUS 2022–2023 ($n = 16,684$ respondents). O1 and O3 are absent from the
home for the majority of peak grid hours (09:00–17:00), while O2 and O4 are
home throughout the day—a 14-fold difference in daytime building presence;
59.8% and 61.4% of O1 and O3 weekday hours in that window are spent in
work activity, against 0.4% and 7.9% for O2 and O4. Within each occupancy
cluster, household composition is varied to capture differences in thermostat
authority and social dynamics: single-person (O1, O4), two-adult couple
(O2), and family with children (O3) [22, 25]. O4 has the smallest sample
($n = 107$); its 15-minute probability tables are the noisiest of the four strata.
Additional strata (multi-adult non-family, student dormitories) would draw
from the remaining 10,643 “other” respondents but would fragment already
thin subsamples without a clearly defined DR participation profile.

Table C.1: ATUS 2022–2023 occupancy metrics and sample sizes by stratum. % working 09–17 is the average weighted probability of being engaged in work activity during peak hours on weekdays (Appendix A). Work location columns are derived from the ATUS TEWHERE variable restricted to employed respondents on diary days with a work episode.

ID	Stratum	% working 09–17 (wkdy)	% at workplace	% at home	n
O1	Employed single adult	59.8	70.2	24.3	1,393
O2	Retired couple	0.4	8.1	50.1	2,351
O3	Employed parent	61.4	68.9	25.7	2,190
O4	Unemployed adult	7.9	0.0	98.4	107

Current executable software version

Table 9: Software metadata (mandatory)

Nr.	Software metadata description	Please fill in this column
S1	Current software version	v1.0.0
S2	Permanent link to executables of this version	https://pypi.org/project/buildocc/
S3	Legal software license	Apache License 2.0
S4	Computing platforms / OS	Linux, macOS, Windows (Python 3.11+)
S5	Installation requirements & dependencies	<code>pip install buildocc</code> ; LLM provider API key required
S6	Link to user manual	https://github.com/humanbuildingsynergy/Build0cc/blob/main/README.md
S7	Support email for questions	wooyoungjung@arizona.edu

References

- [1] T. Hong, S. C. Taylor-Lange, S. D'Oca, D. Yan, S. P. Corgnati, Advances in research and applications of energy-related occupant behavior in buildings, *Energy and Buildings* 116 (2016) 694–702. doi:10.1016/j.enbuild.2015.11.052.
- [2] D. Yan, W. O'Brien, T. Hong, X. Feng, H. B. Gunay, F. Tahmasebi, A. Mahdavi, Occupant behavior modeling for building performance simulation: Current state and future challenges, *Energy and Buildings* 107 (2015) 264–278. doi:10.1016/j.enbuild.2015.08.032.
- [3] M. Arslan, S. Munawar, Large language models in building energy applications: A survey, *Energy & Buildings* 352 (2026) 116800. doi:10.1016/j.enbuild.2025.116800.
- [4] ASHRAE, ANSI/ASHRAE/IES Standard 90.1-2022: Energy standard for buildings except low-rise residential buildings, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA (2022).

- [5] J. Lyu, D. Lai, H. Zhang, Z. Lian, LLM-powered agent for occupant-centric indoor environment control: Design, implementation, and evaluation, *Energy and Built Environment* In press (2026). doi:10.1016/j.enbenv.2026.04.001.
- 710 [6] T. He, F. Jazizadeh, Context-aware LLM-based AI agents for human-centered energy management systems in smart buildings, arXiv:2512.25055 (2025).
URL <https://arxiv.org/abs/2512.25055>
- 715 [7] Z. Qi, Z. Yang, M. Liu, Z. O'Neill, Energy-efficient HVAC control in residential buildings through occupant intention inference with multimodal large language models, *Advances in Applied Energy* 23 (2026) 100283. doi:10.1016/j.adapen.2026.100283.
- [8] W. Jung, Multi-agent home energy management assistant (HEMA), *SoftwareX* 34 (2026) 102633. doi:10.1016/j.softx.2026.102633.
720 URL <https://arxiv.org/abs/2602.15219>
- [9] W. Jung, K. Jeon, P. Babon-Ayeng, What you know does not show: Domain knowledge and occupant-AI interaction in building energy management, in: *Proceedings of the 2026 ACM Sustainability Week*, 2026, pp. 501–508. doi:10.1145/3765611.3815499.
- 725 [10] L. Nicholas, S. Gauthier, E. Gerding, S. Stein, Towards generative occupants: LLM agents in physics-based building simulation, in: *ACM Sustainability Week Companion '26*, Banff, AB, Canada, 2026. doi:10.1145/3765611.3813753.
- [11] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, M. S. Bernstein, Generative agents: Interactive simulacra of human behavior, in: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi:10.1145/3586183.3606763.
730
- [12] M. J. Lee, R. Zhang, LLM-driven generative agents for simulating occupant feedback in built environments, *Journal of Computing in Civil Engineering* 40 (4) (2026) 04026045. doi:10.1061/JCCEE5.CPENG-7365.
735
- [13] A. N. N. Rende, T. Yilmaz, Ö. Ulusoy, Negotiating comfort: Simulating personality-driven LLM agents in shared residential social networks, arXiv:2507.09657 (2025).
URL <https://arxiv.org/abs/2507.09657>

- [14] C. Chen, Ö. Karaduman, X. Kuang, Behavioral generative agents for energy operations, arXiv:2506.12664 (2025).
URL <https://arxiv.org/abs/2506.12664>
- [15] Z. Leng, et al., AgentSense: Virtual sensor data generation using LLM agents in simulated home environments, arXiv:2506.11773, AAAI 2026 (2025).
URL <https://arxiv.org/abs/2506.11773>
- [16] U.S. Bureau of Labor Statistics, American time use survey, 2022–2023 microdata, <https://www.bls.gov/tus/> (2024).
- [17] U.S. Department of Energy, EnergyPlus version 23.2.0, <https://energyplus.net> (2023).
- [18] U.S. Energy Information Administration, Residential energy consumption survey (RECS) 2020, <https://www.eia.gov/consumption/residential/> (2023).
- [19] S. Ramírez, FastAPI: Modern, fast web framework for building APIs with Python, <https://fastapi.tiangolo.com> (2023).
- [20] Anthropic, Model context protocol (MCP) specification, <https://modelcontextprotocol.io> (2024).
- [21] W. Jung, S. Witt, Understanding smart thermostat adoption: Housing, HVAC, and socio-economic traits in the U.S., Energy Reports 15 (2026). doi:10.1016/j.egyr.2026.109244.
- [22] S. Gyamfi, S. Krumdieck, Price, environment and security: Exploring multi-modal motivation in voluntary residential peak demand response, Energy Policy 39 (5) (2011) 2993–3004. doi:10.1016/j.enpol.2011.03.012.
- [23] S. Kim, S. Chang, W. Jung, L. Debs, S. E. Jung, Integrating social justice criteria into multiscale spatial modeling of energy burden, Energy Research & Social Science 134 (2026). doi:10.1016/j.erss.2026.104630.
- [24] W. Jung, Y. Mo, Adoption, use, and perception of room and personal comfort systems in U.S. households, Journal of Building Engineering 127 (2026). doi:10.1016/j.jobe.2026.116313.

- [25] M. H. Albadi, E. F. El-Saadany, A summary of demand response in electricity markets, *Electric Power Systems Research* 78 (11) (2008) 1989–1996. doi:10.1016/j.epsr.2008.04.002.
- 775 [26] W. Jung, P. Babon-Ayeng, Large language model-driven context-aware eco-feedback generation and evaluation, *Energy and Buildings* (2026). doi:10.1016/j.enbuild.2026.118038.
- [27] K. Fisher, J. Gershuny, E. Altintas, A. H. Gauthier, Multinational time use study: User’s guide and documentation, Tech. rep., Centre for Time Use Research, Oxford, UK, <https://www.timeuse.org/mtus> (2013).
- 780 [28] Eurostat, Harmonised European time use surveys — 2018 guidelines, Tech. rep., Publications Office of the European Union, Luxembourg, ISBN 978-92-76-01964-0; <https://doi.org/10.2785/926903> (2019).
- [29] P. O. Fanger, *Thermal Comfort: Analysis and Applications in Environmental Engineering*, Danish Technical Press, Copenhagen, 1970.
- 785 [30] W. Jung, P. Babon-Ayeng, Language as a sensor: Narrative-enhanced thermal comfort modeling, in: *Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 2026. doi:10.1145/3744256.3812583.
- [31] P. Babon-Ayeng, W. Jung, Assessing indoor heat strain through occupant language: A framework for building thermal resilience, *Building and Environment* 304 (2026). doi:10.1016/j.buildenv.2026.115052.
- 790 [32] U.S. Department of Energy, Prototype building models, <https://www.energycodes.gov/prototype-building-models> (2023).
- [33] W. Jung, What LLM-simulated users can and cannot tell us about conversational energy management systems, in: *Proceedings of the 2026 ACM Sustainability Week*, 2026, pp. 428–432. doi:10.1145/3765611.3815484.
- 795 [34] R. El Makroum, S. Zwickl-Bernhard, L. Kranzl, Agentic AI home energy management system: A large language model framework for residential load scheduling, *Results in Engineering* 29 (2026) 109857. doi:10.1016/j.rineng.2026.109857.
- 800 [35] U.S. Department of Energy, The ecobee smart thermostat dataset (2017), <https://www.osti.gov/dataexplorer/biblio/dataset/1854924> (2021).
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Dear Editor:

I am pleased to submit the manuscript "BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research" for consideration in SoftwareX.

BuildOcc is an open-source Python platform that grounds large language model (LLM) agents in the American Time Use Survey (ATUS) — a nationally representative diary dataset covering 16,684 respondents — to simulate building occupant behavior for energy research. Existing occupant behavior models rely on fixed schedules or simple stochastic processes that cannot reason about a person's history, current context, or the type of energy signal being delivered. Prior LLM-agent proposals in this space have used hand-crafted profiles or small convenience samples and have not released a general-purpose, survey-grounded platform for reuse. BuildOcc fills this gap. Each simulated occupant maintains a persona derived from ATUS population statistics, a timestep-level memory stream with periodic reflection, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface (Python library, REST API, and Model Context Protocol server) so that any building energy tool can integrate behavioral intelligence without bespoke coupling code. A plugin registry allows the community to add new demographic strata, custom schedulers, and alternative memory backends as separate installable packages. BuildOcc is an original, openly released software contribution (Apache License 2.0, v1.0.0 at github.com/humanbuildingsynergy/BuildOcc) that addresses a concrete gap in the building energy simulation community:

We believe that this article strongly contributes to the building energy research community. Also, it fits the scope and aim of SoftwareX and complies with its requirements. This manuscript has not been published and is not under consideration for publication elsewhere.

Thank you very much for your consideration.

Sincerely,

A handwritten signature in black ink, appearing to read 'WJung'.

Wooyoung Jung, Ph.D.

Assistant Professor, The University of Arizona

